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GENDER–AWARE CARDIOVASCULAR DISEASE PREDICTION

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Abstract

Cardiovascular diseases remain one of the leading causes of death worldwide. Women often present with different symptoms, risk profiles, and disease progressions compared to men. However, due to the historical underrepresentation of women in clinical studies, most diagnostic and prognostic models have been developed predominantly on male-centric data. This imbalance introduces gender-specific bias into predictive systems.

This research explores and investigates the sources, impact, and mitigation of bias in cardiovascular disease datasets using a combination of machine learning techniques and fairness evaluation. As AI-automated decision-making systems become increasingly common, ensuring fairness and transparency in such models is critical. The study begins by implementing baseline classification using different machine learning models on the dataset, where Logistic Regression and Random Forest showed the highest recall among the evaluated classifiers. This is followed by exploratory data analysis (EDA) to understand and investigate the dataset. Subsequently, feature-specific analysis is performed to detect and identify bias. To address these issues, different mitigation strategies were applied, and the results showed improvement in women's recall. This research aims to contribute to the growing field of fair machine learning in healthcare systems by demonstrating the need for model-specific and context-aware bias mitigation strategies.

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1 Introduction

Cardiovascular diseases (CVDs) remain the leading cause of mortality worldwide, accounting for approximately 17.9 million deaths annually and representing a major global health burden [1]. Early detection and accurate risk prediction are therefore essential for effective prevention and management. Traditional statistical models have long been used for cardiovascular risk assessment; however, they often fail to capture complex nonlinear relationships and interactions among risk factors [2]. These limitations have led to the growing adoption of machine learning (ML) approaches, which leverage large-scale clinical data to improve predictive performance and enable more personalized risk estimation.

Despite these advancements, cardiovascular disease does not manifest uniformly across populations, particularly between men and women. Women are more likely to present with atypical symptoms such as fatigue, nausea, and shortness of breath rather than the classic chest pain commonly observed in men [3]. These differences often lead to delayed diagnosis, misclassification, and poorer clinical outcomes. Furthermore, disparities in healthcare delivery persist, with women being less likely to receive appropriate diagnostic testing and treatment, highlighting systemic gender bias in cardiovascular care [4].

A major contributor to these disparities is the historical underrepresentation of women in cardiovascular research and clinical datasets. Many predictive models have been developed using male-dominated data, limiting their ability to generalize effectively to female populations [5]. As a result, these models often underestimate risk in women and fail to capture sex-specific differences in disease progression and risk factors. Recent studies further emphasize that feature importance and risk factor interactions vary across sexes, suggesting that uniform modeling approaches may not be sufficient [6]. While machine learning models such as ensemble methods and boosting techniques have demonstrated improved predictive accuracy, they are inherently dependent on the quality and distribution of training data. When trained on biased datasets, these models can inherit and amplify existing inequalities, leading to systematic performance disparities across demographic groups [7]. This issue, commonly referred to as algorithmic bias, arises from multiple sources including data imbalance, measurement bias, and structural inequalities embedded within healthcare systems [8]. In cardiovascular prediction tasks, such bias often manifests as higher false negative rates for female patients, indicating that disease cases are more likely to be missed [9].

Bias in machine learning is not confined to model training alone but can occur throughout the entire pipeline, including data collection, preprocessing, and evaluation. In cardiovascular applications, this is further complicated by sex-specific differences in risk factor contributions, which, if ignored, can lead to systematically skewed predictions [1]. Moreover, studies in cardiovascular imaging and other healthcare domains have demonstrated that biased models can result in unequal performance across subgroups, reinforcing disparities in clinical decision-making [10].

To address these challenges, recent research has explored various bias mitigation strategies, including data balancing techniques such as oversampling and synthetic data generation, as well as fairness-aware model training [11, 12]. While these approaches have shown promise, they often involve trade-offs between fairness and predictive performance and may not generalize across different datasets [13]. This highlights the need for more robust and context-specific solutions to ensure equitable model behavior.

In this context, this research aims to investigate gender bias in machine learning models for cardiovascular disease prediction. By analyzing model performance across male and female populations and examining the effects of data imbalance and feature selection, this study seeks to identify the underlying causes of bias and evaluate potential mitigation strategies. Ultimately, the goal is to contribute to the development of fair, reliable, and clinically meaningful predictive models that support equitable healthcare outcomes for all patients.

1.1 Objectives and Research Questions

The main objective of this research is to investigate whether gender bias exists in cardiovascular disease prediction within the selected dataset and to develop strategies to address it. To achieve this, the performance of machine learning models is analyzed separately for male and female patients using key evaluation metrics such as accuracy, precision, recall, and F1–score in order to quantify performance disparities between the two groups. Building on these findings, the study develops and evaluates a range of bias mitigation strategies, including data balancing through synthetic data generation techniques such as SMOTE, gender–specific feature analysis and modeling to address feature–level bias, fairness–aware methods using established toolkits such as Fairlearn, and gender–specific model development. Finally, the effectiveness of these strategies is assessed by examining whether predictive performance, particularly recall for female patients, improves in a way that reduces the gender gap in model outcomes and supports more equitable AI–driven cardiovascular disease prediction.

1.1.1 Problem Statement

Many publicly available healthcare datasets, including cardiovascular disease datasets, exhibit significant class imbalance across demographic attributes, particularly gender. In most cases, male patients are overrepresented, while female samples constitute a relatively smaller portion of the dataset. This imbalance introduces a critical challenge in machine learning model development: models trained on such skewed data tend to learn patterns that are more representative of the majority group, often resulting in suboptimal predictive performance for underrepresented groups, especially women.

In the context of heart disease prediction, this issue is particularly concerning. Clinical evidence suggests that symptoms, risk factors, and disease manifestation differ between males and females, yet standard machine learning pipelines often treat gender as just another feature without addressing underlying distributional bias. As a result, models may achieve high overall accuracy while exhibiting poor recall or sensitivity for female patients, leading to potential underdiagnosis.

Research Questions

To fulfill the objective of this study, the following research questions are addressed:

Research Question 1:

How can gender-based bias be formally defined and measured in cardiovascular disease prediction models with respect to the given dataset?

Research Question 2:

How effective are bias mitigation techniques in reducing gender-based disparities without significantly compromising overall model performance?

1.2 Research Structure

This report is organized into seven chapters to provide a clear structure for investigating and mitigating gender bias in cardiovascular disease prediction. The structure is as follows:

Chapter 01: Introduction

This chapter introduces the topic, presents the motivation, and outlines the objectives and research questions of the study.

Chapter 02: Literature Review and Related Work

This chapter reviews prior work related to bias in datasets, fairness in machine learning, and

existing bias detection and mitigation techniques.

Chapter 03: Methodology
This chapter describes the dataset, preprocessing steps, evaluation metrics, and baseline models used in the study.

Chapter 04: Bias Detection
This chapter presents three different approaches to bias detection, including exploratory data analysis, feature importance analysis, and subfeature importance analysis.

Chapter 05: Bias Mitigation
This chapter presents the different mitigation strategies explored in the study, including data imbalance handling through synthetic data generation, SMOTE, and reweighting, as well as feature-level bias mitigation using Fairlearn and Exponentiated Gradient. It also includes female-specific model development.

Chapter 06: Evaluation
This chapter compares the baseline models with the different bias mitigation approaches and evaluates their effectiveness.

Chapter 07: Future Work and Conclusion
This chapter summarizes the main findings of the study, discusses its limitations, and outlines possible directions for future work.

2 Literature Review

2.1 Background

2.1.1 Bias and Fairness in AI and Machine Learning

Bias and fairness have become central topics in machine learning because automated systems are now used in areas where predictions can influence real decisions and real outcomes [7]. In such settings, good overall performance is not enough on its own. A model may appear accurate at the aggregate level and still perform unevenly across different groups if bias is present in the data, the modeling process, or the evaluation strategy [14]. For this reason, fairness is no longer treated as an optional extension of model development, but as an important condition for building trustworthy and reliable AI systems. Mehrabi et al. describe bias in machine learning as a broad problem that can take different forms and enter at different stages of the pipeline. Bias may arise during data collection, data labeling, feature selection, model development, or evaluation [7]. This shows that unfair model behavior does not always begin at the final prediction stage.

In many cases, the source of the problem is already present earlier in the pipeline, for example, through underrepresentation, missing information, or systematic differences in how groups are recorded. When such issues are carried forward into model training and testing, they can produce unequal outcomes for certain populations even when the model appears technically strong [14]. The literature also makes clear that fairness problems should not be reduced to technical error alone. Danks and London argue that algorithmic bias is not a single uniform issue, since it can arise from training data, algorithmic focus, algorithmic processing, transfer context, and interpretation. Their discussion is useful because it shows that similar unfair outcomes can emerge from different underlying causes [15]. A model may behave unfairly because the data are imbalanced, because the model emphasizes patterns that favor one group, or because predictions are interpreted in a context different from the one in which the model was trained [15]. This means that fairness problems cannot always be solved by changing the model alone. Another important point is that fairness depends not only on how a model is built, but also on how it is evaluated.

If researchers report only overall accuracy or a single average score, important subgroup differ-

ences can remain hidden [14]. A model can therefore look acceptable in a general sense while still performing worse for underrepresented or disadvantaged groups [7]. This is why fairness research pays attention not only to model design, but also to data quality, subgroup-level evaluation, and the real-world context in which the system is used [7, 15].

Taken together, these studies show that bias and fairness should be treated as core questions in machine learning rather than as secondary concerns checked only after a model has been built.

2.1.2 Bias in Healthcare AI and Cardiovascular Applications

The problem of bias becomes even more serious in healthcare, where machine learning models are often trained on clinical data that were not originally collected for fairness analysis. Unlike controlled research datasets, healthcare data are usually produced during routine care. As a result, they reflect not only patient health, but also documentation habits, care pathways, institutional practices, and differences in how patients interact with the healthcare system [8]. This makes healthcare AI especially sensitive to bias, because the model may learn patterns that reflect the way data are generated rather than purely medical relationships. Chen et al. discuss several ways in which bias can enter electronic health record-based systems, including missing data, selective documentation, measurement differences, and temporal changes in clinical practice [8]. These issues do not affect all patients in the same way. Some groups may receive more complete testing, earlier diagnosis, or more detailed documentation, while others may be represented less clearly in the data. In that case, the dataset captures not only disease patterns, but also unequal patterns of observation and treatment. A model trained on such data may then reproduce these differences without distinguishing between clinical reality and data-related distortion [8].

These concerns are also highly relevant in cardiovascular applications, where AI systems are increasingly used for diagnosis, risk prediction, and imaging-based assessment. Cardiovascular AI is often presented as a promising tool for improving early detection and supporting decision-making, but its quality still depends heavily on the data and evaluation strategy behind the model [10]. Van Assen et al. note that bias in cardiovascular AI may result from unbalanced datasets, limited representation of some patient groups, and evaluation practices that do not examine subgroup performance carefully [10]. This matters because strong overall performance can create a misleading impression of reliability even when disparities remain at the subgroup level. In clinical settings, such disparities matter because they may affect later decisions and patient outcomes [8]. If one group is represented less completely or less accurately in the data, the resulting model may perform less reliably for that group, even if the average score looks acceptable [10]. For that reason, fairness in healthcare AI cannot be treated as a small add-on after model development. It needs to be considered from the beginning, including the choice

of data, the assumptions behind preprocessing, and the question of whether all relevant patient groups are being assessed properly [8, 10].

Overall, the literature shows that bias in healthcare AI is not only a technical issue, but also a data and evaluation problem with direct implications for medical decision-making.

2.1.3 Gender Bias in Cardiovascular Disease Diagnosis, Prevention, and Treatment

Gender bias in cardiovascular disease has been discussed across diagnosis, prevention, and treatment [16]. This is especially important because cardiovascular disease has often been understood through a male-centered clinical view, even though women may show different symptoms, risk patterns, and disease progression [16]. In practice, this can influence how symptoms are interpreted, how risk is estimated, and how strongly preventive action is taken. As a result, female patients may face delays in recognition or may not receive the same level of preventive attention and treatment intensity as male patients [3].

Desai et al. explain that sex and gender differences influence cardiovascular risk, presentation, and outcomes [16]. Their argument shows that the issue is not only about fairness in an abstract sense, but also about clinically relevant differences that should matter in prevention and management [16]. If these differences are overlooked, there is a risk that the standard view of cardiovascular disease will mainly reflect male patterns of illness while female-specific patterns remain less visible. This affects both clinical practice and research because the way the disease is defined, studied, and recognized shapes the data that later becomes available for predictive modeling.

Al Hamid et al. similarly report that gender bias appears across diagnosis, prevention, and treatment, which suggests that the problem is not limited to one isolated stage of care [3]. This broader pattern shows how bias can accumulate across the clinical pathway. If women are less likely to be identified early, receive less targeted prevention, and experience different treatment decisions, then the resulting records may reflect an already biased care process [3]. In that sense, bias in cardiovascular care is not only a clinical problem by itself, but also a source of distortion for later data-driven systems. Supporting clinical work has also shown that sex differences in cardiovascular risk are clinically meaningful and may vary depending on the underlying condition and disease context. This supports the view that sex-aware analysis is medically relevant and not only methodologically desirable [4].

When real differences in symptom presentation, risk structure, and disease progression exist, a single general pattern may not describe all patients equally well. Taken together, these studies

show that bias already exists in the cardiovascular domain before machine learning models are applied [16].

This point is especially relevant for the present work, because a model trained on clinical cardiovascular data may learn not only disease-related information, but also patterns shaped by earlier bias in diagnosis, prevention, and treatment.

2.1.4 Sex Disparities in Machine Learning-Based Cardiovascular Prediction

Recent studies have shown that machine learning models for cardiovascular prediction do not always perform equally for male and female patients [13]. This is a serious issue because a model can look accurate in overall evaluation while still producing weaker results for one sex group [5]. In medical settings, such uneven performance can influence risk estimation, diagnosis, and later treatment decisions [9]. For this reason, overall performance alone is not enough when evaluating whether a cardiovascular prediction model is suitable for clinical use. Li et al. show that group-wise evaluation is necessary to understand whether performance differences are hidden by aggregate results [13]. A combined score across all patients can make the model appear stronger than it really is for a subgroup [13].

If male and female patients are not assessed separately, it becomes difficult to identify whether one group benefits more from the model than the other. In this sense, subgroup-based evaluation is not only a fairness exercise but also a basic requirement for understanding model behavior more accurately. Straw et al. report sex-based disparities in cardiac disease prediction algorithms and argue that these differences should be examined directly rather than assuming that one model works equally well for all patients [5]. Their findings show that sex disparity is not only a theoretical concern in fairness research, but something that can be observed in real predictive systems [5]. In healthcare, this matters especially because even moderate differences in sensitivity or error rates can influence downstream decisions. If one group receives less reliable predictions, the model may contribute to unequal care even when it is presented as objective or data-driven.

Harding et al. further show that gender imbalance in training data can influence myocardial infarction prediction [9]. This supports the concern that biased datasets can lead to biased model behavior [9]. The significance of this finding is that it connects performance disparity directly to the composition of the training data rather than only to the model architecture. If one sex group is underrepresented or represented differently in the data, the model may learn patterns that fit one group more strongly than the other. In that case, disparity is not only an outcome observed during evaluation, but something that may already be built into the model during training. At the same time, the literature suggests that these disparities are not explained

only by data imbalance or technical design. Kwak et al. found sex-specific associations between cardiovascular risk factors and atherosclerotic cardiovascular disease, which suggests that the relationship between predictors and outcomes may differ by sex [6].

Sara et al. also discuss sex-specific differences in cardiovascular risk prediction and argue for subgroup analysis when evaluating model behavior [1]. These findings suggest that sex disparity in prediction may reflect both fairness-related problems and meaningful clinical heterogeneity. In other words, weaker performance for female patients may arise partly because women are underrepresented in the data, but also because female cardiovascular risk may not be captured well by models built mainly around dominant patterns.

Wenzl et al. provide further support by showing that an established cardiovascular risk model underestimated risk in women and that a revised approach improved prediction for female patients [2]. This result is especially relevant because it shows that sex disparity is not only detectable but also clinically meaningful. If risk is systematically underestimated in women, delayed recognition or insufficient preventive action may follow. Taken together, these studies suggest that sex disparities in cardiovascular prediction are linked not only to technical bias but also to broader differences in data representation, clinical assumptions, and risk structure [2]. This makes subgroup-aware evaluation a central part of fair and clinically useful model assessment in cardiovascular machine learning.

2.1.5 Bias Mitigation Strategies and Research Gap

Because of these concerns, several studies have examined ways to reduce bias in medical machine learning systems. The literature shows that mitigation is possible, but it is rarely as simple as applying one technical fix and expecting the fairness problem to disappear [17]. In practice, the effect of mitigation depends on the source of the bias, the structure of the data, and the way fairness is evaluated. For that reason, bias mitigation should be understood as a careful part of model development and evaluation rather than as a final correction step. One common direction in the literature is data-level mitigation. Shahul et al. discuss synthetic data generation as one possible strategy for reducing bias in healthcare AI [12]. This approach is useful because it can help address underrepresentation when some patient groups appear less frequently in the available data [12].

Boughareb et al. also show that oversampling methods such as SMOTE can help address class imbalance in medical prediction tasks, including heart disease-related settings [11]. These approaches are relevant because imbalance is one of the most common reasons why models learn majority-group patterns more strongly than minority-group patterns. At the same time, balancing the training data does not automatically guarantee fairer or clinically better predictions. If

the underlying relationships remain weakly captured or if the generated samples do not represent subgroup patterns well, fairness improvements may remain limited [11, 12].

Another important direction is evaluation-focused mitigation. Patricia Cabanillas Silva et al. emphasize that fairness should be assessed through subgroup-based analysis rather than through overall model performance alone [18]. This is especially important in the present context because a mitigation strategy cannot be judged only by whether the average score improves. If the goal is to reduce sex-related disparity, then the evaluation must directly examine whether male and female patients are treated more equally by the model after mitigation [18]. In this sense, subgroup-aware evaluation is not separate from mitigation, but a necessary part of deciding whether mitigation has actually worked. The literature also shows that mitigation should not ignore predictive quality.

Work in biomedical AI suggests that fairness gains should be considered together with model performance, since bias mitigation may involve trade-offs depending on the task and data setting [17]. This matters because a method that improves fairness by severely weakening predictive usefulness may not be practically acceptable in healthcare. The challenge is therefore not only to reduce disparity, but to do so in a way that remains clinically meaningful. This is especially relevant in cardiovascular prediction, where false negatives can have serious consequences and where fairness must be considered together with sensitivity, reliability, and interpretability. Taken together, existing studies show that bias in medical machine learning can be detected and, to some extent, mitigated, but they also show that results depend strongly on the application context [12, 17].

For cardiovascular prediction in particular, the literature still leaves an important gap. There is still a need for clearer evaluation of how sex imbalance affects subgroup-level model performance, how gender-related disparities appear in cardiovascular prediction models trained on imbalanced clinical data, and which mitigation strategies improve fairness without weakening clinical usefulness [5, 13].

This is where the present work is positioned. It focuses on examining gender bias in cardiovascular disease prediction models, evaluating performance separately for male and female patients, and studying how fairness can be assessed and improved in this specific setting.

3 Methodology

This section describes the foundational components underpinning the research framework, including the provenance and structural composition of the dataset, the empirical justification for its selection, and a comprehensive characterization of the clinical attributes employed in the prediction of heart disease. The dataset was systematically evaluated for its suitability in addressing the research objectives, with particular consideration given to its clinical relevance, feature diversity, and representational sufficiency [7, 10]. A comprehensive evaluation of each feature is conducted to illustrate its physiological significance and its potential contribution to the performance of predictive models. Collectively, these elements establish the methodological groundwork upon which the subsequent analytical procedures are built.

3.1 Dataset Description

Cardiovascular diseases (CVDs) are the leading cause of death worldwide, responsible for approximately 17.9 million deaths per year [1]. Heart failure is one of the most serious outcomes of cardiovascular diseases (CVDs). This means that early detection is crucial for ensuring timely treatment and prevention [3, 16]. The Heart Failure Prediction Dataset used in this study contains clinical records of 918 patients and was compiled by combining five publicly available heart disease datasets. Each record includes 11 clinical features and a binary target variable indicating the presence (1) or absence (0) of heart disease. This dataset is well-suited to the development of machine learning models for predicting heart disease risk, thereby supporting early clinical decision–making [1].

3.1.1 Dataset Source

The dataset utilized in this study is the Heart Failure Prediction Dataset, which has been developed to facilitate research on the early detection of cardiovascular disease through the application of machine learning techniques. Each row in the dataset represents a unique patient record, containing clinical attributes along with a binary variable indicating the presence or absence of heart disease. The dataset was created by combining five well-known heart disease datasets and removing duplicate records, resulting in a total of 918 patient observations with 11 clinical features. The dataset is freely accessible and has been extensively utilized in medical prediction and classification research.

Source: The data is sourced from Kaggle's "Heart Failure Prediction Dataset" repository published by Fedesoriano (September 2021) and compiled from multiple heart disease datasets originally available in the UCI Machine Learning Repository and other clinical research sources.

3.1.2 Feature Overview

After removing duplicate records identified across the combined datasets, the final dataset contains 918 unique patient observations. The dataset consists of 11 input features and 1 target variable, where each feature represents a clinically relevant attribute used in cardiovascular risk assessment. In order to facilitate better understanding and analysis, the variables are divided into three categories: numerical features, categorical features and the target variable. A detailed description of all variables and their respective data types is presented in Table 3.1.

Feature Name	**Data Type**	**Description**
Age	Numerical	Age of the patient (years)
Sex	Categorical	Sex of the patient (M: Male, F: Female)
ChestPainType	Categorical	Type of chest pain (TA: Typical Angina, ATA: Atypical Angina, NAP: Non-Anginal Pain, ASY: Asymptomatic)
RestingBP	Numerical	Resting blood pressure (mm Hg)
Cholesterol	Numerical	Serum cholesterol level (mg/dl)
FastingBS	Categorical	Fasting blood sugar (1: if FastingBS $>$ 120 mg/dl, 0: otherwise)
RestingECG	Categorical	Resting electrocardiogram results (Normal, ST, LVH)
MaxHR	Numerical	Maximum heart rate achieved (between 60 and 202)
ExerciseAngina	Categorical	Exercise-induced angina (Y: Yes, N: No)
Oldpeak	Numerical	ST depression induced by exercise relative to rest (numeric value)
ST_Slope	Categorical	Slope of the peak exercise ST segment (Up, Flat, Down)
HeartDisease	Target	Output class (1: Heart disease, 0: Normal)

Table 3.1: Description of Features in the Heart Failure Prediction Dataset

3.2 Data Pre-processing

Effective data preprocessing is a critical step in the machine learning pipeline, directly impacting model accuracy, reliability, and generalizability [7,14]. For the Heart Failure Prediction Dataset, preprocessing ensured data quality, consistency, and proper representation of both numerical and categorical features. Key steps included data inspection, handling of missing values, categorical encoding, and normalization of numerical attributes. These procedures were systematically

implemented using established Python libraries, including Pandas, NumPy, and Scikit-learn, to prepare a structured, model-ready dataset.

3.2.1 Dataset Info

The structure of the dataset was examined using the `.info()` function in Python, which provides an overview of the number of observations, feature names, data types, and the presence of missing values. The output indicates that the dataset contains 918 observations and 12 variables, consisting of 11 input features and one target variable. The information summary also reveals the presence of missing values in certain features, particularly *RestingBP* and *Cholesterol* as shown in Fig. 3.1, which require appropriate pre-processing before Further analysis and model development.

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 918 entries, 0 to 917
Data columns (total 12 columns):
 #   Column          Non-Null Count  Dtype
---  ------          --------------  -----
 0   Age             918 non-null    int64
 1   Sex             918 non-null    object
 2   ChestPainType   918 non-null    object
 3   RestingBP       917 non-null    float64
 4   Cholesterol     746 non-null    float64
 5   FastingBS       918 non-null    int64
 6   RestingECG      918 non-null    object
 7   MaxHR           918 non-null    int64
 8   ExerciseAngina  918 non-null    object
 9   Oldpeak         918 non-null    float64
 10  ST_Slope        918 non-null    object
 11  HeartDisease    918 non-null    int64
dtypes: float64(3), int64(4), object(5)
memory usage: 86.2+ KB
```

Figure 3.1: Output of the `.info()` function showing the structure of the dataset

3.2.2 Data Cleaning

Data cleansing is defined as the systematic process of identifying and rectifying inconsistencies and erroneous entries within a dataset. The purpose of this process is to ensure the accuracy, completeness, and reliability of the data prior to any analytical or predictive modeling procedures [8, 12].

During data preprocessing, both *RestingBP* and *Cholesterol* features were found to contain zero values, which are clinically invalid and indicative of missing measurements. These zero entries were therefore converted to missing values (NaN) to ensure proper handling. To impute

these missing values, a median-based strategy grouped by gender was applied, wherein the median values of *RestingBP* and *Cholesterol* were computed separately for each sex and used for imputation. This approach preserves the inherent physiological differences between male and female populations, maintains realistic feature distributions, and ensures that these variables contribute meaningfully to the predictive performance of the model. As shown in Fig. 3.2, the issue of invalid zero values is effectively resolved after preprocessing.

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 918 entries, 0 to 917
Data columns (total 12 columns):
 #   Column          Non-Null Count  Dtype
---  ------          --------------  -----
 0   Age             918 non-null    int64
 1   Sex             918 non-null    object
 2   ChestPainType   918 non-null    object
 3   RestingBP       918 non-null    float64
 4   Cholesterol     918 non-null    float64
 5   FastingBS       918 non-null    int64
 6   RestingECG      918 non-null    object
 7   MaxHR           918 non-null    int64
 8   ExerciseAngina  918 non-null    object
 9   Oldpeak         918 non-null    float64
 10  ST_Slope        918 non-null    object
 11  HeartDisease    918 non-null    int64
dtypes: float64(3), int64(4), object(5)
memory usage: 86.2+ KB
```

Figure 3.2: Output of the `.info()` function showing the structure after cleaning the dataset

3.2.3 Outlier Detection and Handling

In the context of the Heart Failure Prediction Dataset, outlier detection and handling refer to the systematic identification and treatment of anomalous clinical measurements, such as extreme values in *RestingBP*, *Cholesterol*, *Oldpeak*, and *MaxHR* that deviate significantly from the expected physiological range. Such irregularities can potentially distort model training and adversely impact the accuracy of heart disease predictions [7]. However, in this study, a decision was made to retain outliers, as extreme values may correspond to genuine medical conditions rather than data errors. Preserving these observations ensures that clinically significant cases are not excluded, thereby maintaining the integrity and representativeness of the dataset.

Outlier detection using IQR Method

The Interquartile Range (IQR) method was employed to detect outliers in the numerical features of the dataset. The IQR was calculated as the difference between the 75th percentile (Q3) and

the 25th percentile (Q1). Observations falling below $Q_1 - 1.5 \times IQR$ or above $Q_3 + 1.5 \times IQR$ were identified as potential outliers. These data points were considered statistically anomalous and subjected to further analysis and appropriate handling.

Box plots were used to visualize the distribution and detect outliers in key numerical features, including Age, RestingBP, Cholesterol, MaxHR, and Oldpeak, as shown in Fig. 3.3. Each feature is presented in a separate subplot within a single frame for clear comparison. The plots highlight the median and the interquartile range (IQR), which represents the spread between the first and third quartiles, along with potential outliers beyond this range. This analysis confirms that preprocessing has removed unrealistic values and resulted in more consistent and meaningful feature distributions.

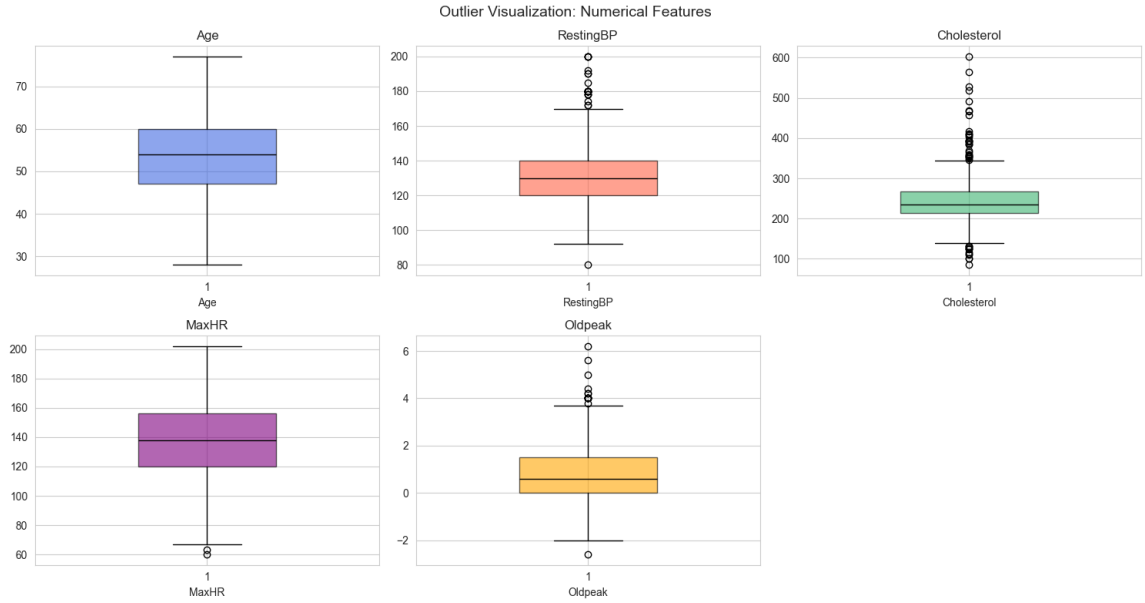


Figure 3.3: Outlier detection for basic numerical features

Key Observations:

- Significant outliers are observed in *Cholesterol*, *RestingBP* and *Oldpeak*, indicating high variability and the presence of extreme clinical cases.
- Moderate outliers in *MaxHR* suggest abnormal physiological conditions, while *Age* remains relatively stable with minimal outliers.

Outlier handling through Winsorization

In order to mitigate the adverse influence of detected outliers while preserving the overall sample size and distributional structure of the dataset, Winsorization was applied as the primary

outlier treatment strategy. In this method, extreme values beyond the defined IQR-based thresholds are capped and replaced with the nearest acceptable boundary values at the 1st and 99th percentiles. This limits the distorting effect of anomalous clinical measurements without discarding potentially informative patient records [11]. The effect of Winsorization on the feature distributions is illustrated in Fig. 3.4 performed on Logistic Regression

	Accuracy	Precision	Recall	F1-score
Before Winsorization	0.842391	0.847619	0.872549	0.859903
After Winsorization	0.842391	0.841121	0.882353	0.861244
Change (After - Before)	0.000000	-0.006498	0.009804	0.001341

Figure 3.4: Effect of Winsorization on classification performance metrics.

Although Winsorization was implemented to mitigate the influence of extreme values, it resulted in only marginal changes in model performance as shown in Fig. 3.4. Consequently, this approach was not adopted further, and the original data distribution was preserved.

These outliers are likely to represent genuine medical conditions rather than noise; therefore, retaining them preserves clinically important information for heart disease prediction.

3.2.4 Feature Transformation

Feature transformation aims to reduce the skewness and tries to improve the distribution of numerical features of a dataset. The goal of feature transformation is to minimize the negative effect of skewness, and by doing so, it enhances the model's performance. Some machine learning models, like logistic regression and SVM, are sensitive towards normality [7]. This process ensures that all features conform to the distributional assumptions, scale requirements, and structural prerequisites of machine learning algorithms. The outcome of this process is enhanced model convergence, predictive stability, and overall analytical interpretability.

3.2.4.1 Feature Scaling

Feature scaling standardizes numerical features to a consistent range for improved model performance, while encoding converts categorical variables into a numerical format that machine learning algorithms can interpret without introducing bias [14].

Standard Scaling and Robust Scaling

Standard scaling transforms numerical features to have zero mean and unit variance. In contrast, robust scaling utilizes the median and interquartile range (IQR), making it less sensitive to extreme values. For the Heart Failure Prediction Dataset, robust scaling was implemented

using the `RobustScaler()` method, as it effectively reduces the impact of outliers, leading to a more stable and representative feature distribution for model training.

Encoding

One-hot encoding was applied to the categorical features of the Heart Failure Prediction dataset, including *Sex*, *ChestPainType*, *RestingECG*, *ExerciseAngina*, and *ST_Slope*, using `pd.get_dummies`. Each categorical variable was transformed into multiple binary columns corresponding to its unique categories; for features with more than two categories, separate dummy variables were created for each category. This approach eliminates any implicit ordinality and ensures that categorical variables are appropriately represented for supervised machine learning algorithms.

Shapiro-Wilk-Test

To ensure whether transformation is really needed, this study also performed the 'Shapiro-Wilk test' method as shown in Fig. 3.5 to see the p-value, for numerical characteristics.

```
Feature         | P-Value                          | Result
-------------------------------------------------------------
Age             | 5.437913633601987e-06            | NOT Normal ✗
RestingBP       | 6.224677214777964e-08            | NOT Normal ✗
Cholesterol     | 1.0310289193855276e-10           | NOT Normal ✗
MaxHR           | 2.679017151570236e-07            | NOT Normal ✗
Oldpeak         | 3.0219016934901185e-27           | NOT Normal ✗

5 out of 5 features are non-normal
```

Figure 3.5: Normality Test Results for Key Features

Key Observations:

- The normality test results in Fig. 3.5 indicate that all numerical features (*Age*, *RestingBP*, *Cholesterol*, *MaxHR*, and *Oldpeak*) have p-values significantly less than 0.05, leading to the rejection of the null hypothesis of normality. This confirms that none of the features follow a normal distribution and are likely skewed or contain extreme values.

To improve the prediction of such models that are sensitive towards skewness, the Yeo-Johnson power-transformation technique was implemented using the `PowerTransformer` method as shown in Fig. 3.6. This transformation stabilizes variance and reshapes feature distributions towards normality without requiring strictly positive values. The resulting transformed features exhibit a centered distribution with reduced skewness, thereby improving model robustness and performance, particularly for algorithms sensitive to feature distribution [8].

```
Sample of transformed values:
        Age  RestingBP  Cholesterol     MaxHR   Oldpeak
0 -1.402913   0.512000     0.994036  1.431777 -0.974838
1 -0.524125   1.486202    -1.385143  0.756825  0.491685
2 -1.677163  -0.052279     0.886499 -1.517776 -0.974838
3 -0.626012   0.403874    -0.527261 -1.145270  0.862390
4  0.002584   1.021694    -0.990062 -0.610564 -0.974838

Post-transformation statistics:
           Age  RestingBP  Cholesterol    MaxHR  Oldpeak
count  918.000    918.000      918.000  918.000  918.000
mean     0.000     -0.000        0.000    0.000   -0.000
std      1.001      1.001        1.001    1.001    1.001
min     -2.113     -2.349       -2.972   -2.263   -2.747
25%     -0.727     -0.683       -0.527   -0.688   -0.975
50%      0.003     -0.052       -0.079    0.020    0.077
75%      0.675      0.512        0.589    0.757    0.862
max      2.389      2.306        2.691    1.996    1.802
```

Figure 3.6: Yeo–Johnson transformation for Key Features

Key Observations:

- The Yeo–Johnson transformation results show that the numerical features are rescaled to a more normalized distribution, with values centered around zero and reduced variance. This transformation effectively addresses the non-normality observed earlier by reducing skewness and stabilizing feature distributions.
- Such pre–processing enhances model performance, particularly for linear models, by ensuring that features contribute more uniformly during training while preserving important clinical information.

3.3 Evaluation Metrics

3.3.1 Fairness Metrics

Demographic Parity Metrics

Demographic parity, also known as statistical parity, is a fairness metric used to assess whether a machine learning model's positive predictions are independent of sensitive attributes such as *Sex* [7]. It evaluates whether different demographic groups receive similar favorable outcomes, thereby helping identify potential bias. In this context, it measures whether the model's prediction of heart disease is dependent on gender.

Equalized Odds

Equalized Odds is a fairness metric that requires a predictive model to achieve similar true positive rates (TPR) and false positive rates (FPR) across different demographic groups defined by a sensitive attribute, such as *Sex* [7]. This criterion ensures that the model's ability to correctly identify positive cases, as well as its tendency to produce false alarms, is balanced across groups.
Formally, Equalized Odds is satisfied when the prediction $\hat{Y}$ is conditionally independent of the sensitive attribute A, given the true outcome Y, i.e.,

$$P(\hat{Y} = 1 \mid A = a, Y = y) = P(\hat{Y} = 1 \mid A = b, Y = y), \quad \forall a, b,\ y \in \{0, 1\}.$$

In the context of heart disease prediction, this implies that the model should exhibit comparable sensitivity (recall) and false positive rates for both male and female patients. This is particularly important in clinical settings, where disparities in error rates across demographic groups may lead to unequal healthcare outcomes [3, 10]. By enforcing Equalized Odds, the model aims to ensure that no group is systematically disadvantaged in terms of missed diagnoses or incorrect predictions.

Equal Opportunity Metrics

Equal Opportunity is a fairness metric that evaluates whether a predictive model achieves equal true positive rates (TPR) across different demographic groups defined by a sensitive attribute, such as *Sex* [7]. It ensures that all groups have an equal chance of being correctly identified when the actual outcome is positive.
Formally, Equal Opportunity is satisfied when:

$$P(\hat{Y} = 1 \mid A = a, Y = 1) = P(\hat{Y} = 1 \mid A = b, Y = 1), \quad \forall a, b.$$

In the context of heart disease prediction, this metric ensures that the model is equally effective in identifying patients with heart disease across male and female groups. The Equal Opportunity gap measures the difference in TPR between groups, where a smaller gap indicates a fairer model.

3.3.2 ML Model Performance Metrics

These metrics evaluate the ML model's outcome based on its effectiveness and accurate predictions for heart disease. The evaluation shows how well a machine learning model can predict [7].

Accuracy

One of the fundamental evaluation metrics is accuracy. Accuracy increases as a model increases the number of correct predictions. So this evaluation sees the amount of correctly predicted values as a proportion of the total number of values. The equation for accuracy is,

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \tag{3.1}$$

Here,
TP = True Positive,
TN = True Negative,
FP = False Positive,
and FN = False Negative.

Precision

Precision is also a very popular evaluation metric for machine learning models, which is especially used in classification problems where the cost of a false positive is high. Precision is the ratio of accurately predicted positive observations (True Positives) to all predicted positives (True Positives + False Positives). So in other words, precision looks at the values which are predicted as true and were supposed to be in the true domain, the equation for precision is,

$$Precision = \frac{TP}{TP + FP} \tag{3.2}$$

Here,
TP = True Positive,
FP = False Positive.

Recall

Recall is also another popular evaluation metric used to evaluate machine learning models. Recall is calculated based on the actual positive cases, which are predicted to be true in proportion to the total correct predictions. Recall is particularly important in medical data analysis, as it measures the ability of a model to correctly identify patients who truly have the disease [3, 16]. The equation for recall is,

$$Recall = \frac{TP}{TP + FN} \tag{3.3}$$

Here,
TP = True Positive,
FN = False Negative.

F1-Score

F1-score is an evaluation metric that actually exposes the balance of precision and recall. F1-score is the harmonic mean of precision and recall. The equation for F1–Score is,

$$F1\text{-}score = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{3.4}$$

3.4 Baseline Model

3.4.1 Experimental Setup

Environment and Toolkit

All the experiments were implemented using Python in Visual Studio Code using the following libraries and tools:

- **Pandas** and **NumPy** for data loading and manipulation.
- **Matplotlib** and **Seaborn** for data visualization.
- **Scikit-learn** for data preprocessing, data splitting, model development, and model evaluation.
- **Fairlearn** for fairness evaluation and bias mitigation.

Dataset and Dataset Splitting:

The dataset used for this study was described in the Chapter [3] where various patients with features such as *Age, Sex, ChestPainType, RestingBP, FastingBS* etc. and the target variable as *HeartDisease* feature.

For evaluation purposes, the dataset was divided into testing and training sets, with an 80% training and 20% for testing split. The split ratio was maintained throughout all the phases of the experiment. In this preliminary phase, multiple classification machine learning models were built and implemented to create a baseline understanding for evaluating predictive fairness and performance metrics. The objective of this phase is to assess how traditional machine learning models perform on the heart disease dataset without applying bias-mitigating and fairness-aware strategies.

3.4.2 Models Implemented

The following machine learning models were implemented based on the findings of the literature review and their use for classification problem [13, 18]. As this section focuses only on the

implementation of machine learning models and their result evaluation, the information and the process of handling missing and duplicate data, data splitting, and evaluation criteria are provided in details in the Section 3.3.

3.4.2.1 Logistic Regression

The first model that was implemented is Logistic Regression, as it achieves a good balance of accuracy and interpretability for a binary classification problem. The model was trained on the full set of features during training, following standard pre-processing steps. As the values of categorical features are either integer or binary, the main preprocessing step was achieved for scaling the numerical features. In this phase, no bias mitigation and feature selection techniques were incorporated during implementation

The model achieved 89% recall, shown in Table 3.2 are the predictions on the test set,

Model Performance

Model	Accuracy	Recall	Precision	F1-score
Logistic Regression	0.89	0.89	0.90	0.90

Table 3.2: Performance metrics of Logistic Regression model

3.4.2.2 Random Forest

Random Forest classifier was implemented in this phase as it combines multiple decision trees with default parameters and aggregates their predictions to improve the decision accuracy. This model was implemented to increase the robustness and to address whether there is any complex pattern in decision making that is consistent with its effectiveness in medical prediction tasks [13].

The model achieved 91% recall, shown in Table 3.3 are the predictions on the test set.

Model Performance

Model	Accuracy	Recall	Precision	F1-score
Random Forest	0.88	0.91	0.88	0.89

Table 3.3: Performance metrics of Random Forest model

3.4.2.3 Gradient Boosting

The Gradient Boosting classifier was implemented to make predictions as it is an ensemble-based algorithm that builds models sequentially by combining multiple weak learners to minimize prediction errors. Gradient Boosting is sensitive to hyperparameters, feature quality, and overfitting, particularly when dealing with noisy or imbalanced data. Therefore, appropriate pre-processing techniques were applied before implementing the model.

The model achieved 86% recall, as shown in Table 3.4, which are the predictions on the test set.

Model Performance

Model	Accuracy	Recall	Precision	F1-score
Gradient Boosting	0.86	0.86	0.89	0.88

Table 3.4: Performance metrics of the Gradient Boosting model

3.4.2.4 Support Vector Machine(SVM)

The Support Vector Machine classifier was implemented using a linear kernel (using the SVC class) as it is one of the most powerful supervised classification algorithms. Since SVM is also sensitive to scaling, all numerical values were standardized during pre-processing phase.

The model achieved 87% recall, as shown in Table 3.5, which are the predictions on the test set.

Model Performance

Model	Accuracy	Recall	Precision	F1-score
SVM	0.85	0.87	0.86	0.87

Table 3.5: Performance metrics of SVM model

3.4.2.5 Decision Tree

The Decision Tree classifier was implemented to identify if there is any pattern in heart disease prediction. The Decision Tree model is one of the most useful models in the baseline phase, as it works well on both numerical and categorical features.

The model achieved 82% recall, shown in Table 3.6 are the predictions on the test set.

Model Performance

Model	Accuracy	Recall	Precision	F1-score
Decision Tree	0.80	0.82	0.82	0.82

Table 3.6: Performance metrics of the Decision Tree model

Result and Key Observation

Overall accuracy and performance metrics of the five implemented models are shown in the following Table 3.7 for a balanced view of model predictions and effectiveness.

Model	Accuracy	Recall	Precision	F1-score
Logistic Regression	0.89	0.89	0.90	0.90
Random Forest	0.88	0.91	0.88	0.89
Gradient Boosting	0.86	0.86	0.89	0.88
SVM	0.85	0.87	0.86	0.87
Decision Tree	0.80	0.82	0.82	0.82

Table 3.7: Performance metrics of baseline models

- Random Forest achieves the best overall performance, outperforming all models with the highest recall (0.91), while Logistic Regression also demonstrates strong baseline performance with high recall (0.89)and balanced metrics, making it a reliable and interpretable reference model.
- SVM, Gradient Boosting, and Decision Tree exhibit moderate results, with SVM performing better than Gradient Boosting and Decision Tree, but still not matching the effectiveness of ensemble methods like Random Forest.

The findings from the baseline models emphasize the need for bias mitigation techniques to ensure reliable prediction of heart disease across demographic groups.

3.4.3 Performance Analysis of 'Sex' in Baseline Models

The performance analysis across gender groups reveals a consistent disparity between male and female predictions in the baseline models, as shown in Table 3.8. For Logistic Regression, male patients achieve high recall and precision (0.92 each), whereas female patients show significantly lower recall (0.50) and precision (0.60), indicating a reduced ability to correctly identify positive cases among females.

Similar trends are observed across other models. Random Forest and Gradient Boosting show relatively better performance for females (recall: 0.67), but still lag behind male recall (0.93 and 0.88, respectively). SVM achieves a higher female recall (0.83), yet female precision remains lower than that of males. The Decision Tree model exhibits the most pronounced imbalance, with very low precision (0.36) and F1-score (0.47) for females. Overall, these results indicate that all baseline models exhibit gender-based performance disparities, with consistently lower recall, precision, and F1-score for female patients. This suggests a systematic bias toward the majority group, leading to poorer detection of heart disease cases in females.

Model	Gender	Accuracy	Recall	Precision	F1-score
Logistic Regression	Female	0.87	0.50	0.60	0.55
	Male	0.89	0.92	0.92	0.92
Random Forest	Female	0.92	0.67	0.80	0.73
	Male	0.87	0.93	0.88	0.90
Gradient Boosting	Female	0.89	0.67	0.67	0.67
	Male	0.89	0.88	0.90	0.89
SVM	Female	0.89	0.83	0.62	0.71
	Male	0.84	0.88	0.88	0.88
Decision Tree	Female	0.76	0.67	0.36	0.47
	Male	0.81	0.83	0.87	0.85

Table 3.8: Gender-wise performance metrics of baseline models

Although Random Forest demonstrates comparatively more balanced performance, with higher accuracy and recall for females (0.92 and 0.67, respectively), a noticeable disparity still exists when compared to male performance (recall: 0.93). Across other models, this gap is more pronounced, with consistently lower recall, precision, and F1-score for female patients. Overall, these results indicate that baseline models tend to favor male instances, leading to poorer predictive performance for female patients. This disparity highlights the presence of gender-based bias in the dataset and justifies the need for fairness-aware mitigation techniques to achieve more equitable model performance.

4 Bias Detection

This chapter presents a holistic bias detection framework using both technical and non–technical approaches applied to the dataset. The goal of this chapter is to find and identify any bias pattern using multiple detection frameworks across 918 patients, enabling robust detection of discrimination.

In recent years, the growing use of machine learning models in healthcare decision-making has highlighted the importance of assessing fairness and identifying potential biases, particularly those related to demographic attributes such as *Sex* [3, 10]. This section systematically evaluates the analytical components of this study to detect patterns, quantify bias, and provide empirical evidence of its presence within the dataset. Furthermore, it examines how such biases may influence model predictions, potentially leading to inequitable outcomes in heart disease diagnosis.

Bias detection and fairness analysis are a crucial part of understanding the dataset and its integrity, impact, and influence on machine learning models. As historical inequalities, imbalanced data representation, and unaddressed discriminatory sensitive features can introduce bias into training data, and machine learning models trained on such data may produce unfair and discriminatory outcomes [7, 14]. So, exploring and identifying the sensitive features, correlation between features, dependencies between features, group-wise analysis, and comparison between features will reveal whether any inequality or imbalance is present in the dataset.

4.1 Exploratory Data Analysis

Exploratory Data Analysis (EDA) constitutes a pivotal phase in the machine learning process, encompassing the examination of dataset structure, distributions, and feature relationships prior to the development of a model. In the Heart Failure Prediction Dataset, the EDA was performed to understand how clinical features vary across patients with and without heart disease. The analysis employed a range of descriptive statistics and visualization methods, including histograms, bar graphs, and box plots. This analysis is instrumental in identifying patterns, correlations, and potential biases in the data [7]. Moreover, it provides guidance on decisions regarding pre-processing, feature selection, and model configuration.

4.1.1 Univariate Analysis

Univariate analysis examines each feature individually to understand its statistical properties without considering relationships with other variables. In the Heart Failure Prediction dataset, the model was applied to numerical as well as categorical features. Visualizations such as histograms and bar plots were employed for numerical and categorical features, as shown in Fig. 4.1 and Fig. 4.2, respectively. This analysis helped in identifying distribution patterns, skewness, and potential data quality issues prior to developing a model.

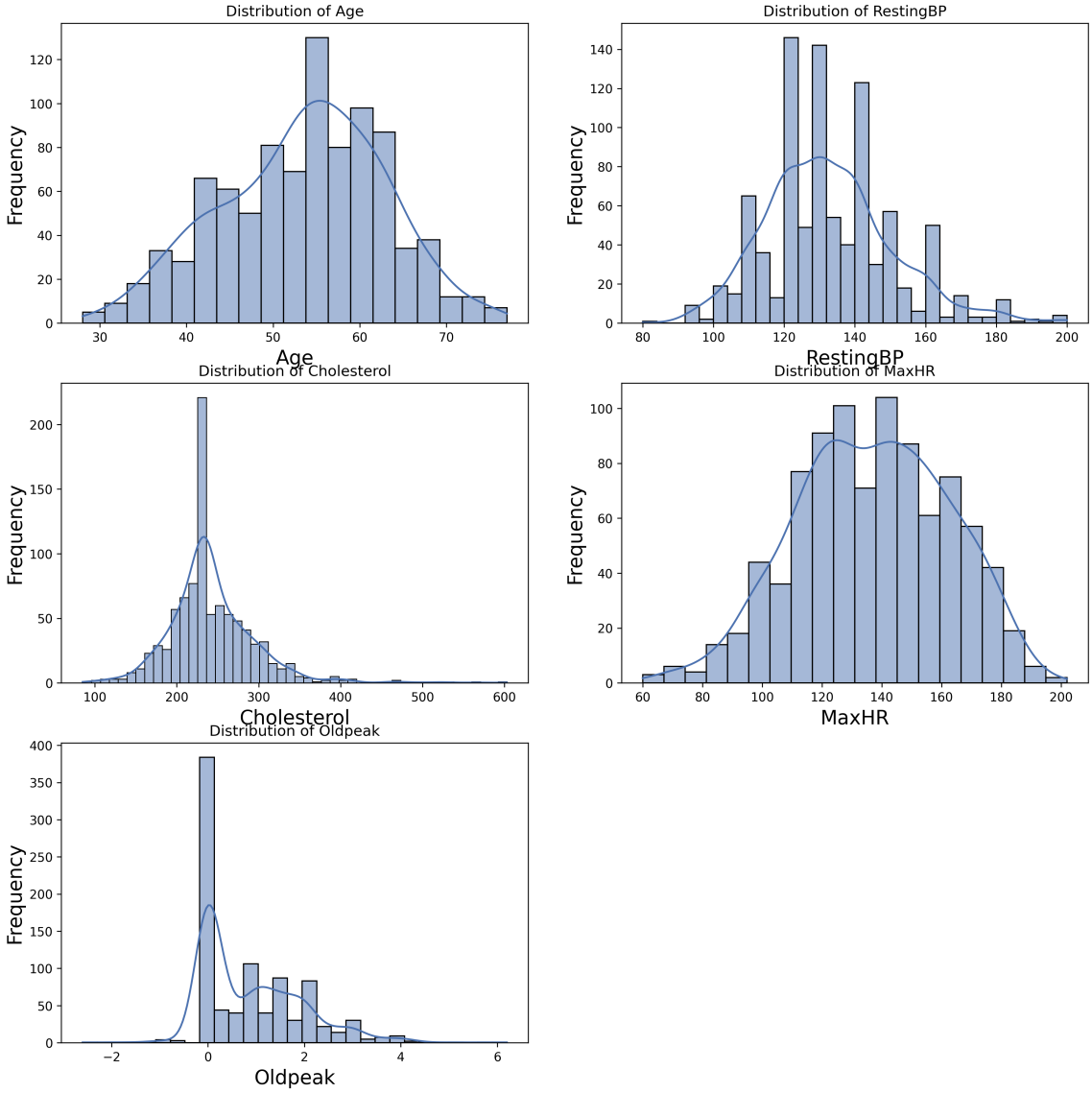


Figure 4.1: Uni–variate Distributions of Numerical Features

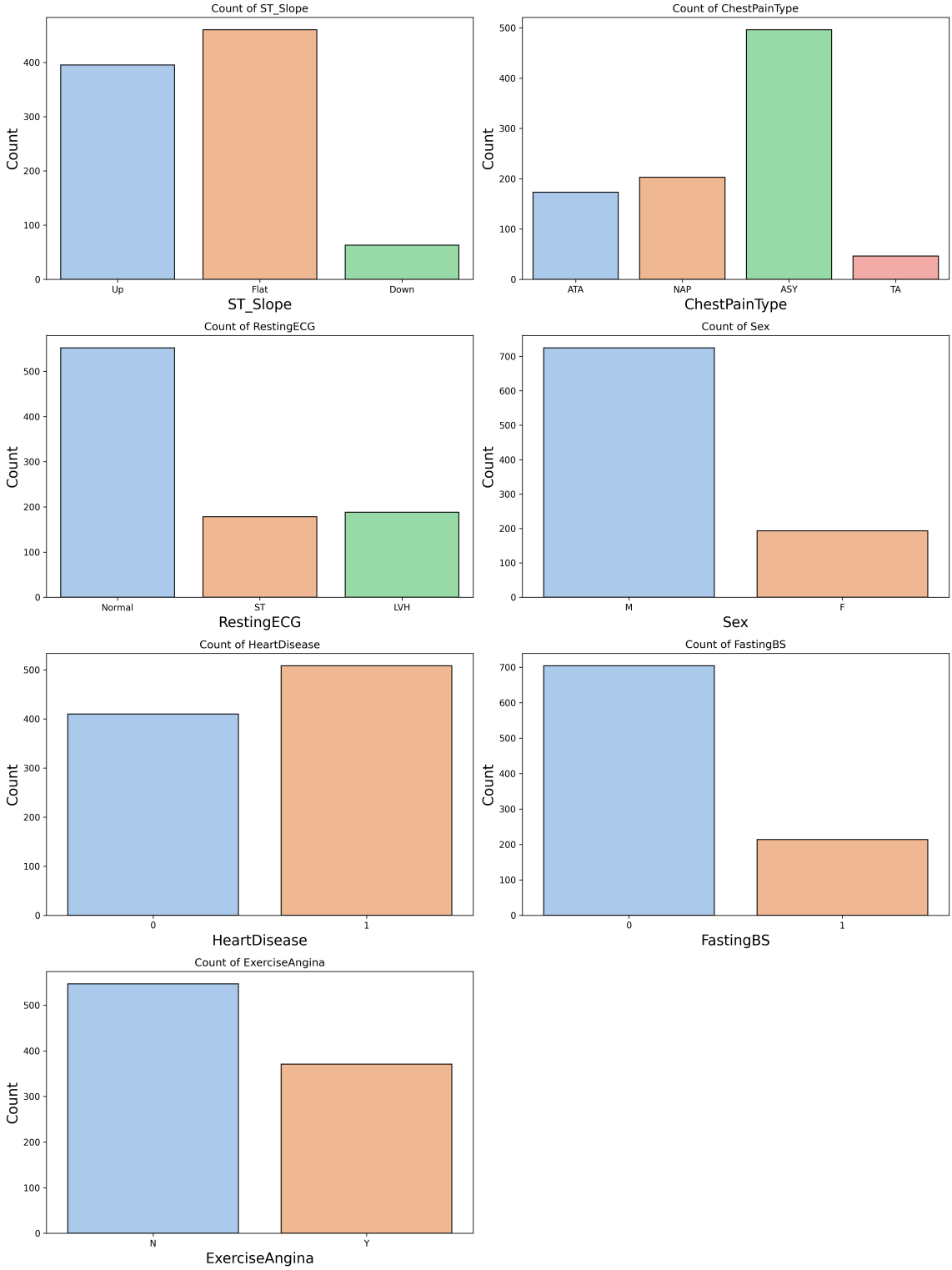


Figure 4.2: Uni–variate Distributions of Categorical Features

Key findings from Uni-variate Analysis

The following Table 4.1. shows the summarized findings of the univariate analysis of each feature.

Feature	**Distribution Shape**	**Observation**
Age	Approximately normal (bell–shaped)	Majority within 45–65 age range, indicating a middle-aged dominant population with slight symmetry and minimal skewness.
RestingBP	Slightly right–skewed	Majority cluster around 110–140 mmHg, with some higher values extending the tail, suggesting presence of mild outliers.
Cholesterol	Highly right–skewed	Most values are between 200–300, but extremely high values indicate significant outliers and variability.
MaxHR	Approximately normal	Symmetrically distributed around 120–160, indicating a well-balanced feature with minimal skewness.
Oldpeak	Strongly right–skewed	High concentration near 0 with a long right tail, indicating low ST depression for most patients.
ST_Slope	Imbalanced	Most patients show flat or upsloping ST segments, while downsloping cases are rare.
ChestPainType	Highly imbalanced	A large proportion of patients are asymptomatic, which is clinically significant and may affect diagnosis.
RestingECG	Dominated by "Normal"	Most patients exhibit normal ECG results, with fewer abnormal cases (ST, LVH).
Sex	Strongly Imbalanced	Dataset is male-dominated, indicating potential gender bias [5, 9].
FastingBS	Highly skewed toward $0 (< 120\,\text{mg/dL})$	Majority of patients have normal fasting blood sugar levels, with fewer high-risk cases.
ExerciseAngina	Slightly Imbalanced	Most patients do not experience exercise-induced angina, though a notable portion does.
HeartDisease	Slightly Imbalanced	About 500 patients with heart disease vs 400 patients without heart disease.

Table 4.1: Summary of uni-variate Analysis – Numerical and Categorical Features vs. Heart Disease

4.1.2 Bi–variate Analysis

The primary focus of Bi-variate analysis is to find the relationship between each feature and the target feature, especially Fig. 4.3 and Fig. 4.4 show the relationship of numerical and categorical features with the target feature HeartDisease, respectively. This analysis methodically elucidates the correlation between physiological characteristics to the development of heart disease [4, 6].

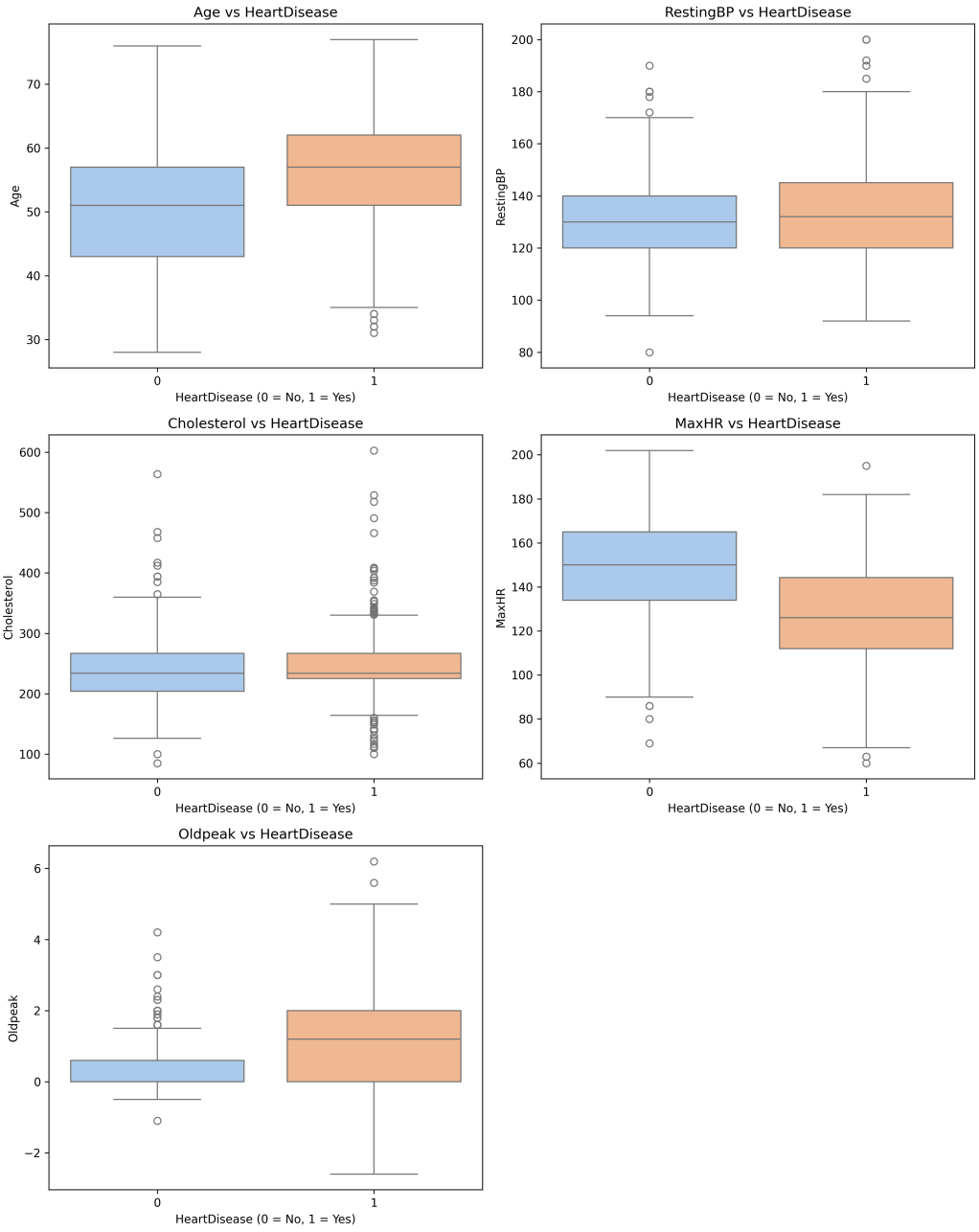


Figure 4.3: Bi-variate distribution between Numerical features vs HeartDisease

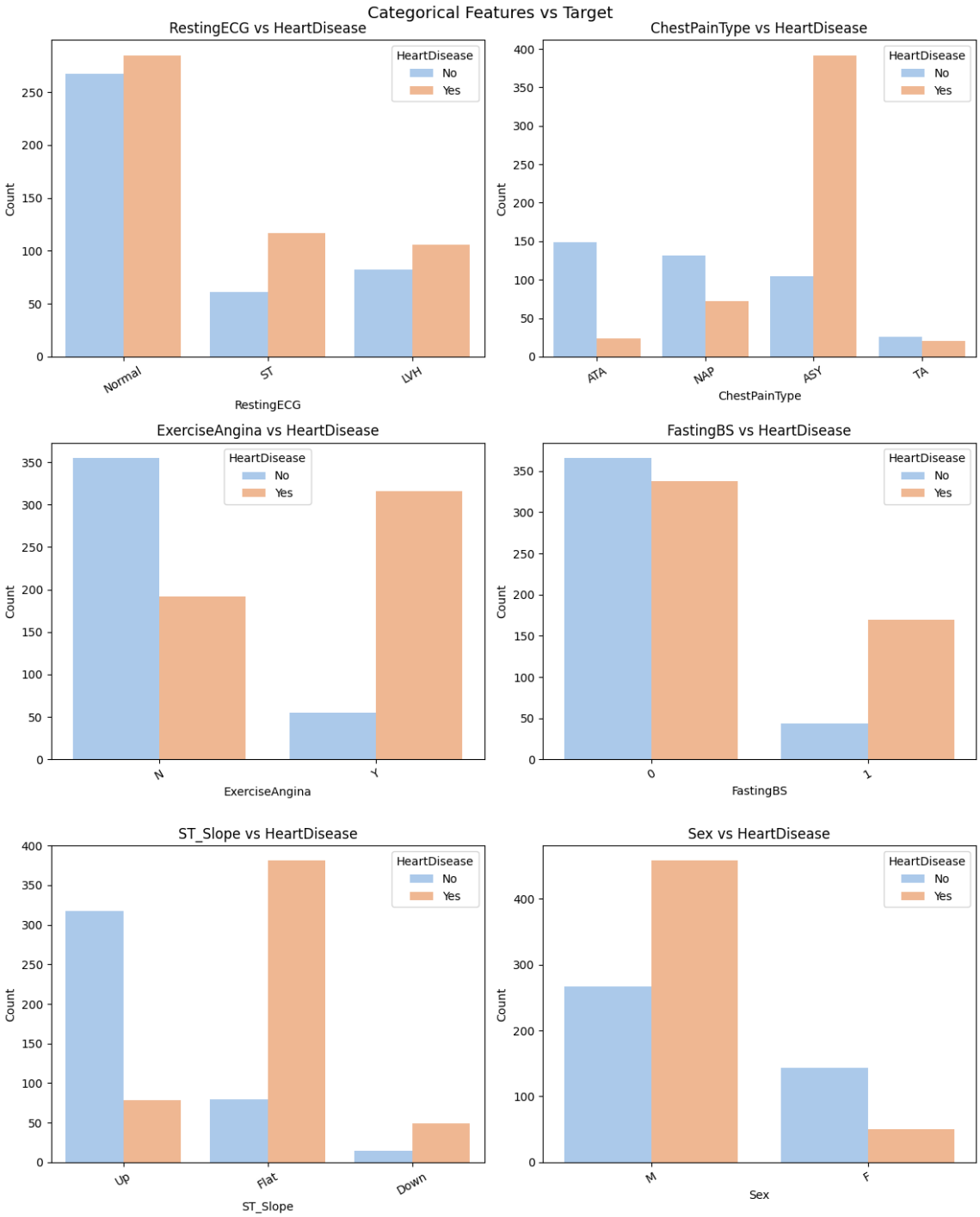


Figure 4.4: Bi-variate distribution between Categorical features vs HeartDisease

Key findings from Bi-variate Analysis

The following Table 4.2. shows the summarized findings of univariate analysis of each feature.

Feature	Observation	Interpretation
Age	Slightly higher median for heart disease	Older individuals are more likely to have heart disease.
RestingBP	Similar distribution across classes	Limited impact on heart disease prediction.
Cholesterol	High overlap with many outliers	Weak predictor despite variability.
MaxHR	Lower values for heart disease patients	Lower heart rate associated with higher disease risk.
Oldpeak	Significantly higher for heart disease	Strong indicator of cardiac abnormality.
ST_Slope	Flat is more frequent in disease cases	Abnormal ST slope strongly linked to heart disease.
ChestPainType	ASY dominant in disease cases	Asymptomatic patients have higher disease risk relative to others.
RestingECG	Dominated by "Normal" and similar for the rest	Weak discriminative feature.
Sex	More males in the disease group	Gender imbalance and bias [5].
FastingBS	Higher proportion of $1(> 120\,\mathrm{mg/dL})$ in disease cases	High blood sugar linked to heart disease.
ExerciseAngina	Much higher in disease cases	Strong predictor of heart disease.

Table 4.2: Summary of Bi–variate Analysis – Numerical and Categorical Features vs. Heart Disease

4.1.3 Group Disparity

Group disparity analysis is a method of evaluating differences in outcomes across demographic subgroups. It is employed to detect potential bias in the dataset or predictive model [7, 18]. In the Heart Failure Prediction Dataset, a comparison was made between heart disease prevalence rate against different demographic subgroups of every categorical feature as shown in Fig. 4.5. The analysis also examined whether clinical features differed significantly across these groups. This approach is intended to ensure fairness and equity in cardiovascular risk prediction [10].

While numerical features were primarily analyzed in their continuous form, the feature *Age* was discretized into meaningful intervals to facilitate group-based disparity analysis as shown in Fig. 4.6. This transformation allows for interpretable comparisons across age groups. Other numerical features were not discretized, as they do not inherently represent demographic groupings and are more appropriately utilized in their continuous form for predictive modeling.

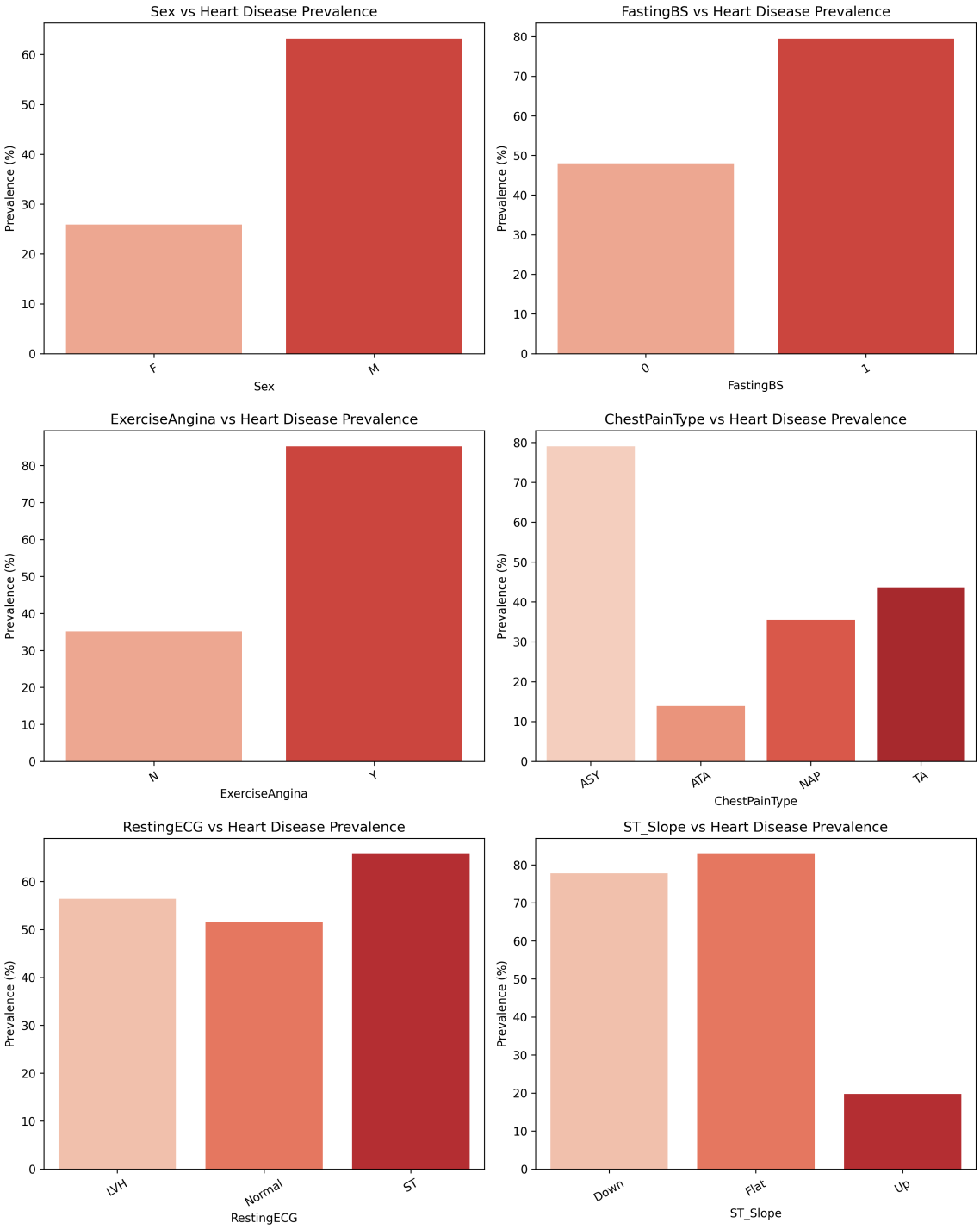


Figure 4.5: Group Disparity between subgroups of Categorical Features

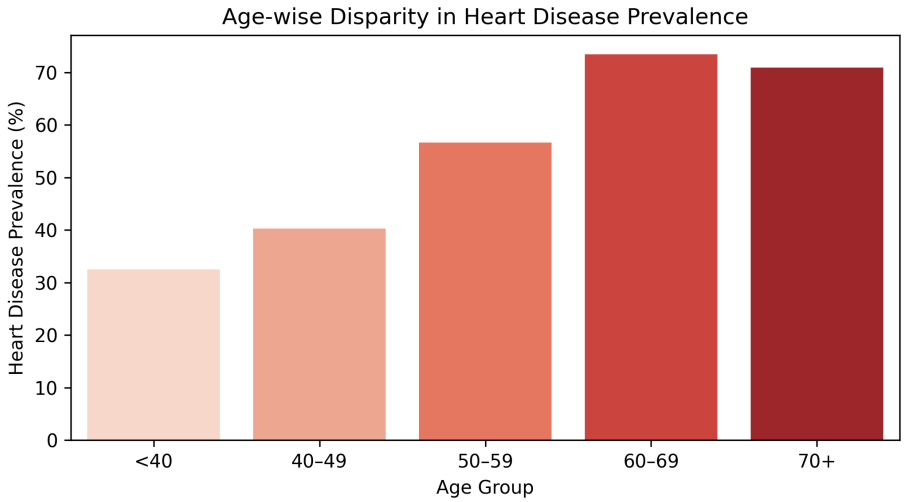


Figure 4.6: Group Disparity of Age

Key Observations:

- A substantial gender disparity is observed from the attribute *Sex*, where heart disease prevalence among males (around 63%) is more than double that of females (around 26%), indicating strong representation and outcome bias in the dataset.
- *ExerciseAngina* shows the strongest discriminatory effect, with prevalence increasing from 35% (No) to 85% (Yes), highlighting its high predictive importance.
- *ST_Slope* demonstrates clear separation, with "Flat" (83%) and "Down" (78%) slopes strongly associated with higher prevalence compared to "Up" (20%), making it a critical feature.
- *ChestPainType* reveals significant variation, where "ASY" (79%) has the highest prevalence, while "ATA" (14%) shows the lowest, indicating strong class differentiation.
- *FastingBS* indicates moderate discrimination, with elevated levels (> 120) showing higher prevalence (79%) compared to normal levels (48%).
- *RestingECG* exhibits relatively weak differentiation, with smaller variation across categories (Normal 52%, LVH 56%, ST 66%), suggesting limited predictive contribution.
- Age-wise analysis shows a clear increasing trend, with prevalence rising from 32% (¡40) to 73% (60–69) and remaining high (70%) for 70+, confirming age as a critical risk factor.
- The magnitude of disparities across *Sex* and key clinical features highlights the need for fairness-aware modeling to prevent biased predictions and ensure equitable performance across demographic groups.

Importantly, the substantial disparity observed in *Sex* indicates potential bias in the dataset, emphasizing the need for fairness-aware modeling and bias mitigation techniques. These findings collectively justify the subsequent application of fairness metrics and bias reduction strategies to ensure equitable and reliable model performance across demographic groups.

4.1.4 Correlation and Dependency Analysis

Correlation analysis is a crucial step in data analysis, as it helps identify relationships between features and detect potential proxy variables that may indirectly influence model predictions [7]. In this study, correlation and dependency analysis are performed to examine relationships between clinical features, the sensitive attribute *Sex*, and the target variable *HeartDisease*.

The objective is to identify potential sources of bias by analyzing both direct and indirect associations between features and the prediction outcome. Clinical variables such as *ST_Slope*, *ChestPainType*, and *ExerciseAngina* may strongly influence predictions, while certain features may act as proxy variables for sensitive attributes, potentially introducing unintended bias.

To capture both linear and non-linear dependencies, techniques such as correlation analysis, mutual information, and Cramér's V are employed. These methods provide insights into feature interactions and help ensure that the model does not rely on biased or indirectly discriminatory patterns, thereby supporting fair and reliable heart disease prediction [7].

Mutual Information Analysis

The primary objective of mutual information analysis is to evaluate the extent to which numerical features contribute to heart disease prediction by capturing both linear and non-linear dependencies [13]. Fig. 4.7 presents the mutual information scores for the numerical features within the dataset.

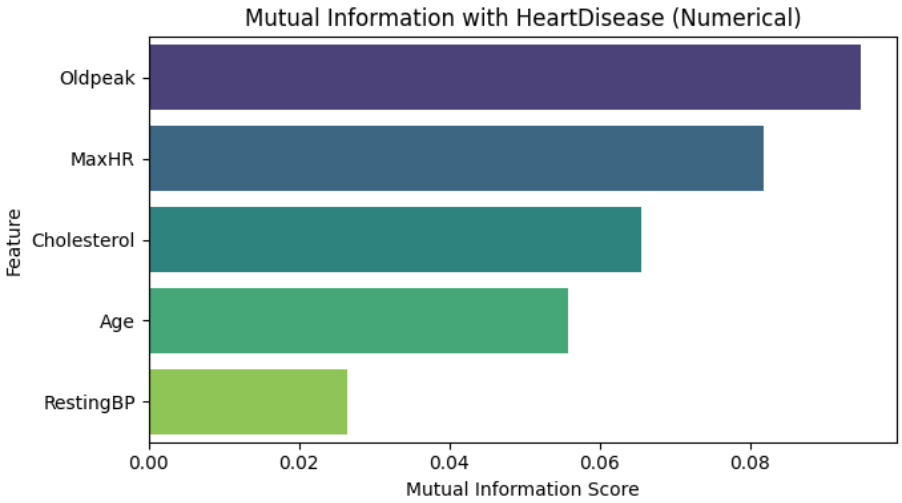


Figure 4.7: Mutual Information analysis for numerical features

Key Observations:

- *Oldpeak* and *MaxHR* are the most informative features, showing the highest contribution to heart disease prediction.
- *RestingBP* has the lowest importance, indicating weak predictive influence compared to other numerical features.

Cramér's V for Categorical Associations

Cramér's V for Categorical associations was computed as shown in Fig. 4.8 to explore the relationship between categorical variables and heart disease prediction [13].

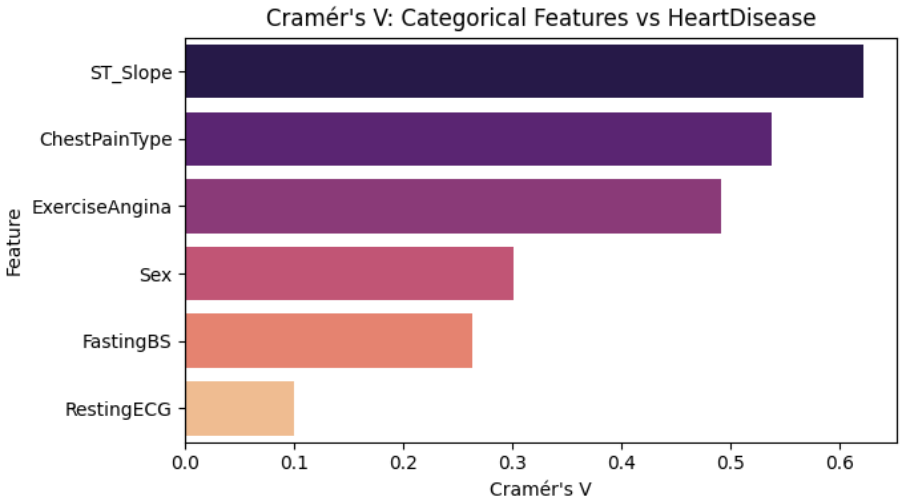


Figure 4.8: Cramér's V for Categorical features

Key Observations:

- *ST_Slope* and *ChestPainType* show the strongest associations with heart disease, indicating high predictive relevance among categorical features.
- *Sex* exhibits a moderate association, suggesting potential influence on predictions and the need for fairness analysis, while *RestingECG* shows the weakest association.

Correlation and Dependency Analysis Summary

In this section, relationships and dependencies between features were examined to identify potential sources of bias. Using correlation-based analysis and mutual information, it was observed that certain numerical features such as *RestingBP* and *Age* have relatively lower influence on heart disease prediction. In contrast, *Oldpeak* and *MaxHR* were identified as the most influential numerical features based on mutual information scores. Furthermore, Cramér's V analysis revealed that categorical features such as *ST_Slope*, *ChestPainType*, and *ExerciseAngina* have strong associations with the target variable *HeartDisease*. The sensitive attribute *Sex* was found to have a moderate association with the target, indicating potential bias in the dataset. These findings highlight the need for further fairness analysis and the application of bias mitigation techniques to ensure equitable and reliable model performance.

4.2 Feature Importance

The feature importance technique is utilized to evaluate and categorize the extent to which each input feature contributes to a machine learning model's predictions [13, 18]. In the Heart Failure Prediction Dataset, this analysis was used to ascertain the impact of clinical variables on heart disease classification as shown in Fig. 4.9 and Fig. 4.10. This analysis elucidates the model's decision-making process and identifies the most predictive attributes. This enables a clear comparison of each feature's contribution to the prediction process.

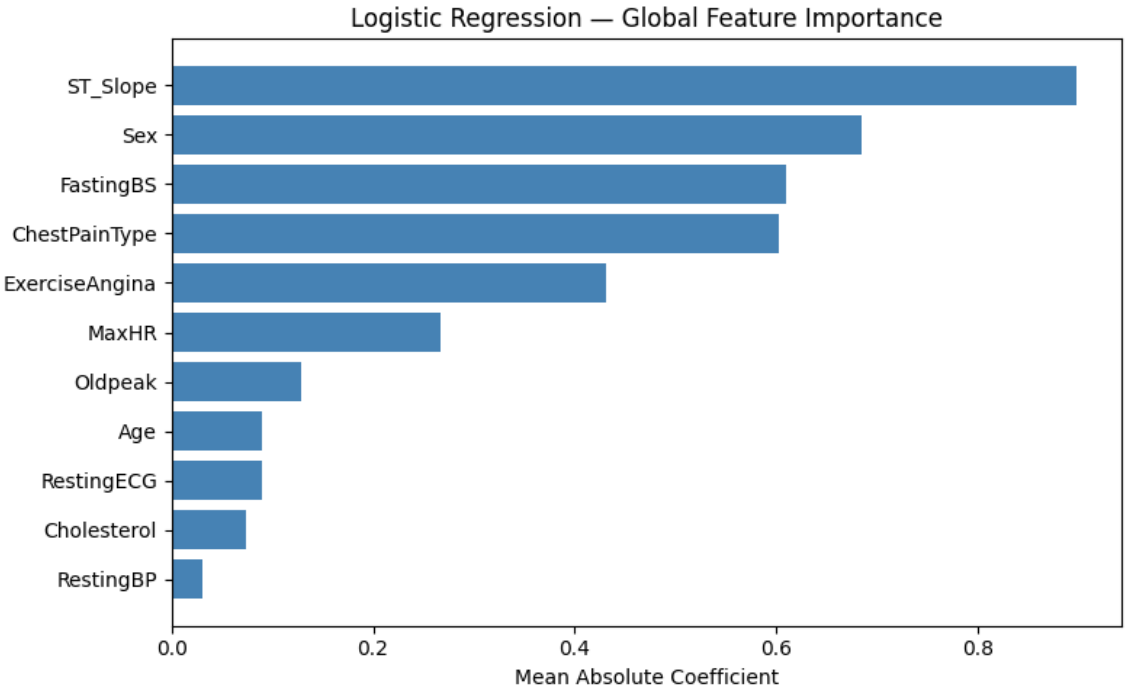


Figure 4.9: Feature Importance Comparison using Logistic Regression

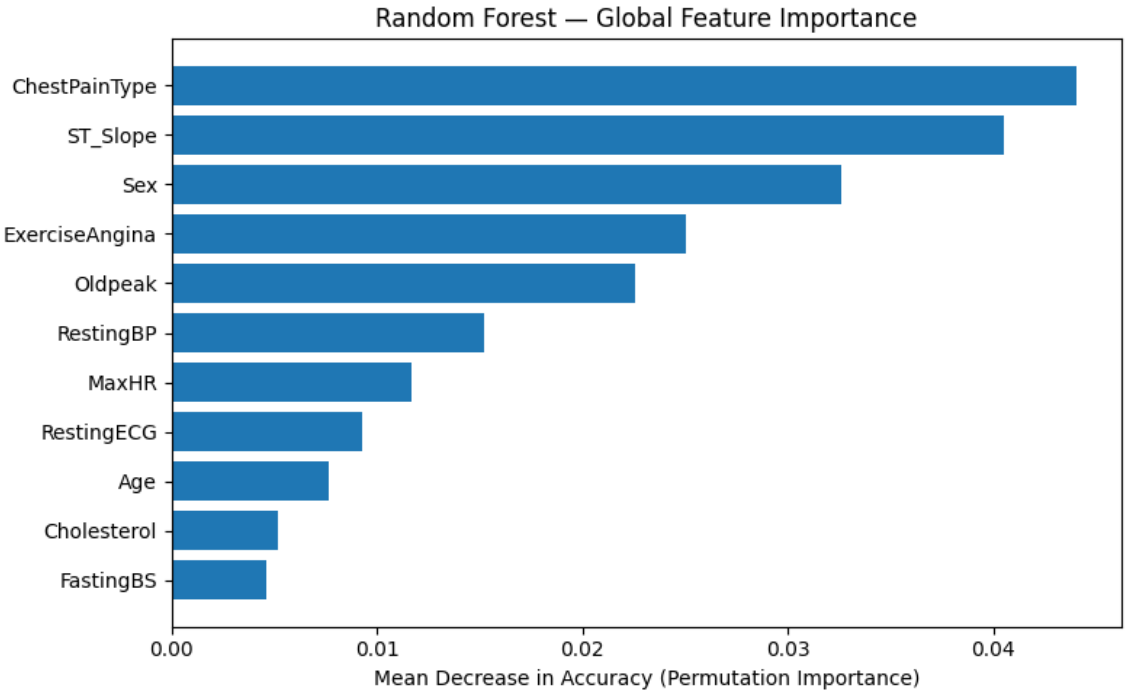


Figure 4.10: Feature Importance Comparison using Random Forest

Key Observations:

- Logistic Regression identifies *ST_Slope* as the most influential feature, followed by *Sex*, *FastingBS*, and *ChestPainType*, indicating strong linear contributions from both clinical and demographic variables.
- Random Forest highlights *ChestPainType* as the dominant feature, followed by *ST_Slope* and *Sex*, demonstrating the importance of clinical symptoms and their non-linear interactions.
- Features such as *ExerciseAngina* and *Oldpeak* show consistent moderate importance across both models, reinforcing their predictive relevance. However, *MaxHR* is moderately important in Logistic Regression but less influential in Random Forest.
- *Sex* remains one of the top features in both models, indicating strong predictive contribution; however, this does not necessarily imply bias but highlights its association with the target variable.
- *FastingBS* shows high importance in Logistic Regression but relatively low importance in Random Forest, suggesting its effect may be more linear and less interaction-driven.
- *Age*, *RestingBP*, and *Cholesterol* exhibit consistently low importance across both models, indicating limited standalone predictive power.
- The variation in feature rankings between the models suggests that Random Forest captures non-linear relationships and feature interactions, whereas Logistic Regression relies on direct linear associations.
- Overall, *ST_Slope*, *ChestPainType*, and *Sex* emerge as robust predictors across models, supporting their clinical significance in heart disease prediction.

Furthermore, the program has been designed considering Sex as a sensitive feature and thereafter removing sex from the feature set to prevent direct bias influence. The models were then trained separately on male and female subsets, and feature importance was computed independently for each group. As illustrated in Fig. 4.11 and Fig. 4.12, noticeable differences in feature contributions can be observed across genders. In both models, ST_Slope and ChestPainType emerge as dominant predictors; however, their relative importance varies between male and female populations. Additionally, other features exhibit differing levels of influence, indicating that the underlying predictive patterns are not uniform across groups. These variations highlight the presence of potential gender-specific biases and justify the need for subgroup-aware modeling approaches.

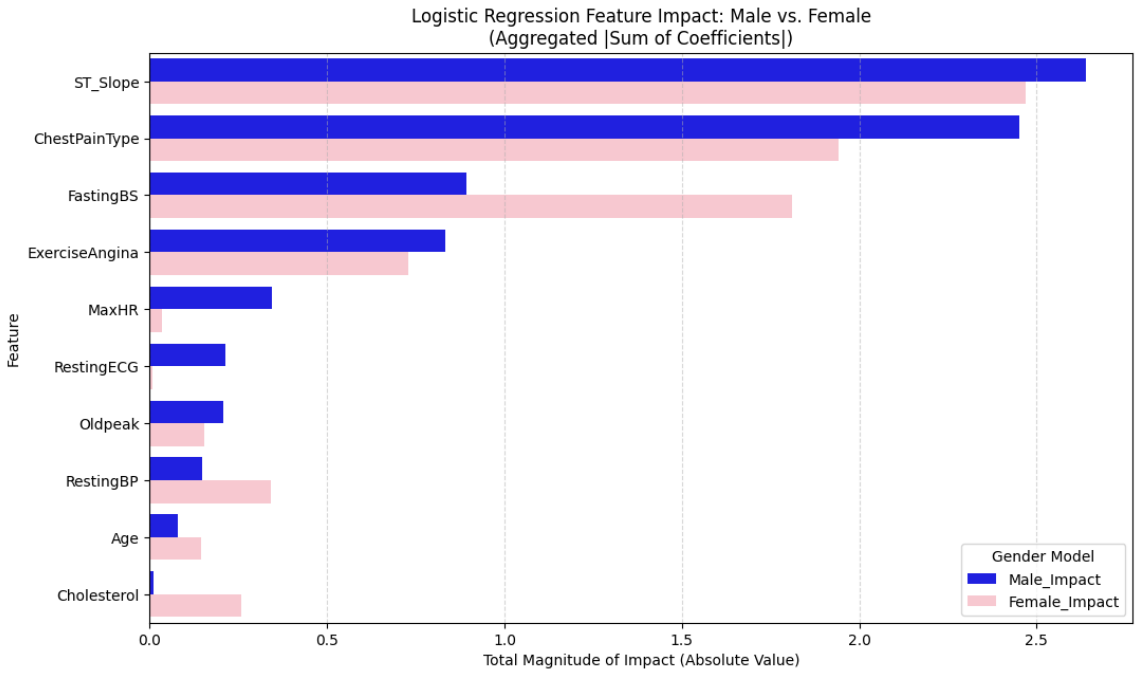


Figure 4.11: Gender Specific Feature Importance - Logistic Regression

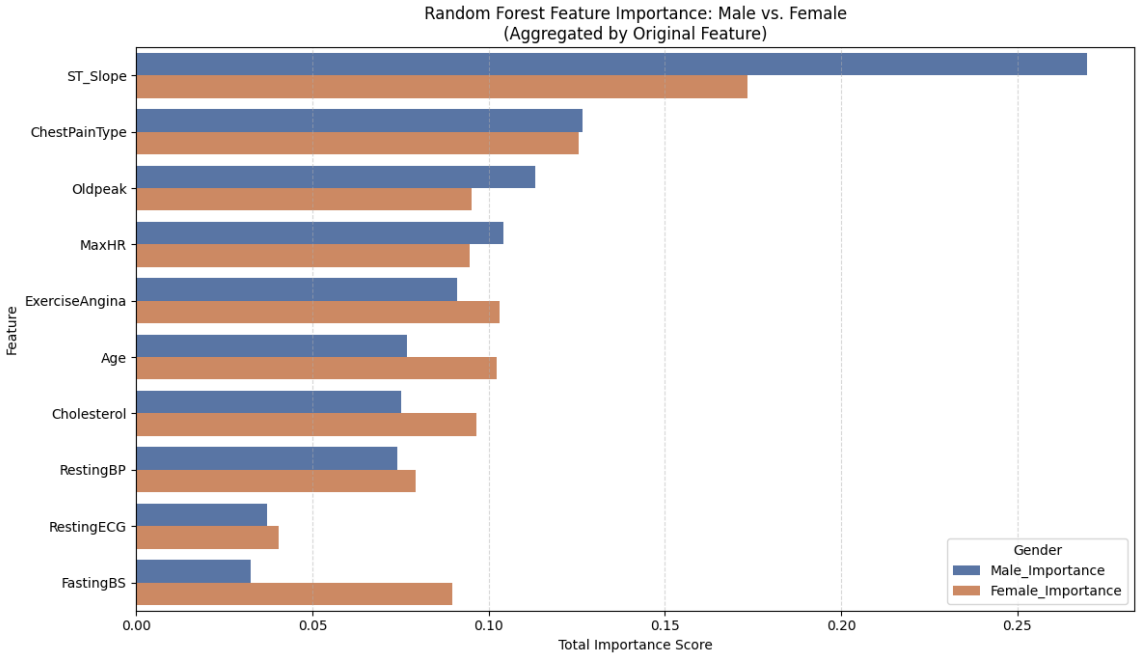


Figure 4.12: Gender Specific Feature Importance - Random Forest

Key Observations:

- *ST_Slope* and *ChestPainType* are the most dominant features across both models and genders, with slightly higher impact observed for males.
- *FastingBS*, *Cholesterol*, and *Age* show relatively higher importance for females in both the models, indicating stronger metabolic and age-related risk patterns in female patients.
- *Oldpeak* and *MaxHR* contribute more prominently for males, particularly in Random Forest, indicating stronger cardiovascular signal representation for male patients.
- *FastingBS* and *RestingBP* show higher importance for females across both models, with the disparity especially pronounced in Logistic Regression.
- *ExerciseAngina* exhibits relatively balanced importance across genders with minor variations, while *RestingECG* remains consistently low in importance for both.
- Overall, the variation in feature importance across genders highlights subgroup-specific patterns, suggesting that models capture different risk profiles for male and female patients, which may introduce bias if not addressed.

4.3 Sub feature Importance

Sub-feature importance is an extension of feature importance that evaluates the predictive contribution of individual encoded categories rather than entire features, which enhances interpretability and fairness evaluation [7, 18]. The focus of this study is on categorical values produced through preprocessing methods, including one-hot encoding. In the Heart Failure Prediction Dataset, the present analysis examined encoded variables from features such as *ChestPainType, ST_Slope,FastingBS, RestingECG, and ExerciseAngina*. analyzed separately. As illustrated in Fig. 4.13 and Fig. 4.14, the specific attribute values that exert the most significant influence on heart disease predictions for both the models, Logistic regression and Random Forest.

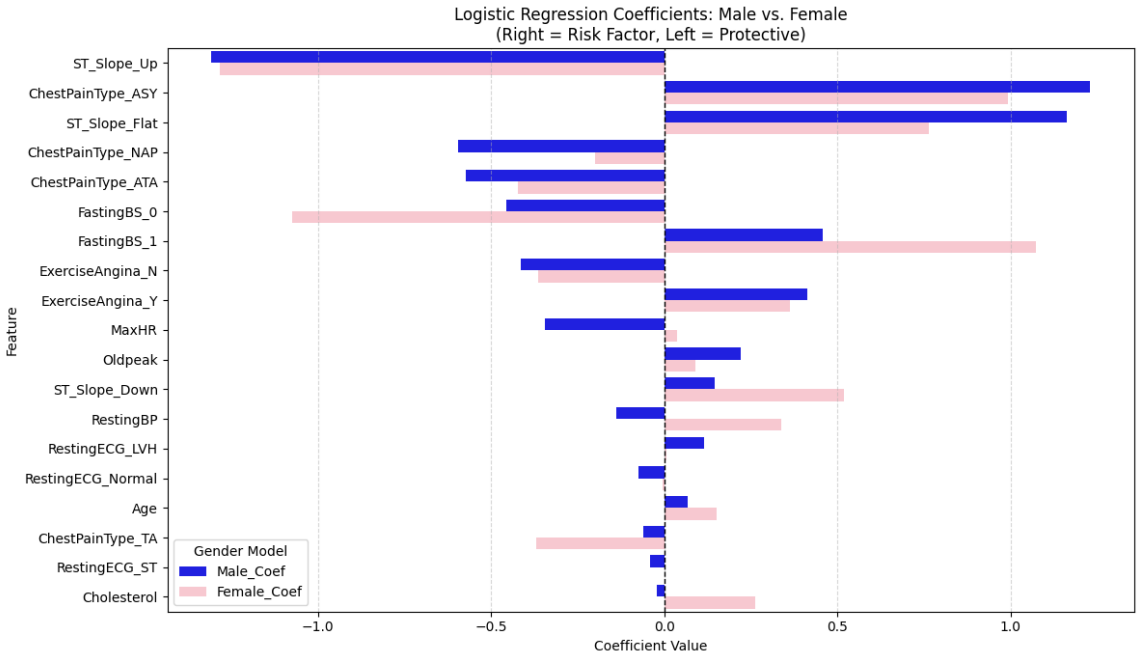


Figure 4.13: Subfeature level Feature Specific Importance - Logistic Regression

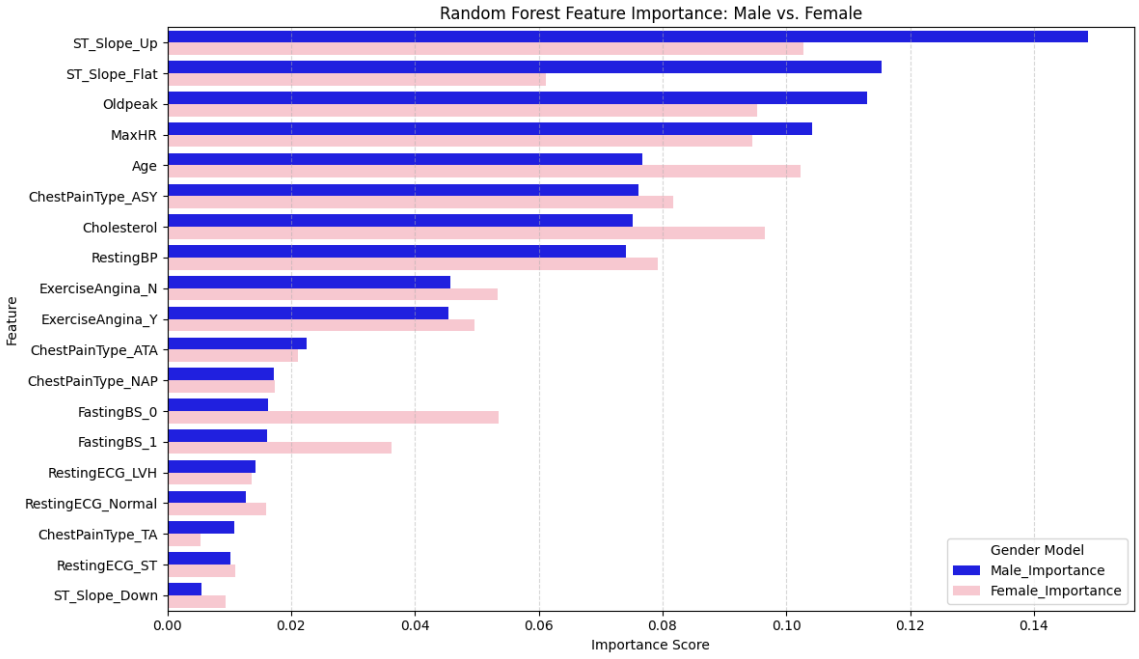


Figure 4.14: Subfeature level Feature Specific Importance - Random Forest

Key Observations:

- *ST_Slope_Up* and *ST_Slope_Flat* are the strongest predictors across both models, with consistently higher importance for males, indicating dominant cardiovascular signal contribution, while *ST_Slope_Down* shows a stronger risk association for females in Logistic Regression.
- *FastingBS_1* exhibits a substantially stronger positive effect for females in Logistic Regression and higher importance in Random Forest, while *FastingBS_0* acts as a strong protective factor, especially for females, highlighting pronounced metabolic risk differences.
- *ChestPainType_ASY* is a strong risk factor for both genders, whereas *ChestPainType_ATA* and *NAP* act as protective factors; however, Random Forest assigns comparatively higher importance to *ChestPainType_ASY* for females, indicating stronger symptom-driven risk patterns.
- *Oldpeak* and *MaxHR* show higher importance for males in Random Forest, suggesting stronger non-linear cardiovascular effects, while *ExerciseAngina* exhibits slightly higher importance for females across both models.
- *Age* and *Cholesterol* contribute more prominently for females in Random Forest and show positive associations in Logistic Regression, indicating their relevance in female-specific risk modeling.
- *RestingBP* demonstrates a stronger positive association for females in Logistic Regression and slightly higher importance in Random Forest, whereas *RestingECG* features consistently exhibit low importance across both models.
- Overall, the differences in coefficient direction and feature importance between male and female groups reveal distinct subgroup-specific risk patterns, suggesting that models capture gender-dependent relationships that may introduce bias if not addressed through fairness-aware approaches.

This approach has been demonstrated to yield more precise insights than the feature-level importance alone. The identification of such findings may suggest the presence of potential discriminatory patterns within the model. This analysis has been shown to enhance the interpretability and fairness evaluation of the predictive system.

5 Bias Mitigation

Building upon the findings from the bias detection techniques, this chapter focuses on addressing feature-level bias in heart disease prediction, with particular emphasis on the sensitive attribute *Sex*, while also considering potential bias arising from other influential clinical features. Bias across demographic groups and feature interactions can lead to unequal predictive performance [7, 14], making it essential to apply targeted mitigation strategies.

Accordingly, this chapter investigates bias associated with *Sex* alongside feature-level dependencies and implements fairness-aware approaches to reduce their impact on model predictions. These techniques aim to ensure that the model achieves more balanced and equitable performance across demographic groups [8, 13], while maintaining robust and clinically meaningful predictions.

5.1 Sensitive Attribute Imbalance Mitigation

Imbalanced representation of demographic groups poses a significant challenge in supervised machine learning, as it can influence both model performance and fairness evaluation. In the context of the Heart Failure Prediction dataset, the distribution of the sensitive attribute *Sex* was found to be highly skewed during univariate analysis, with a predominance of male instances. This imbalance may lead the model to learn patterns biased toward the majority group, potentially resulting in reduced predictive performance and fairness for underrepresented female patients [5, 7, 18].

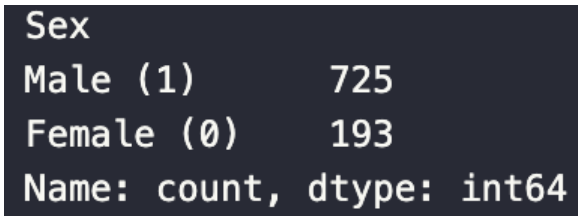


Figure 5.1: Imbalance of Sex Variable

Fig. 5.1 illustrates that the distribution of the sensitive attribute *Sex* is highly imbalanced, with male instances (725, 79%) significantly outnumbering female instances (193, 21%). Such disproportionate representation can lead to biased model behavior, where the classifier predom-

inantly learns patterns from the majority group while underrepresenting the minority group. This imbalance not only affects model generalization but also introduces challenges in fairness evaluation, as performance metrics may be skewed toward the majority group [9, 16].

To address this issue, this phase focuses on analyzing and mitigating bias associated with the *Sex* attribute using fairness-aware techniques [3, 8], aiming to achieve more balanced and reliable model performance across demographic groups. To handle the imbalance in the sensitive attribute, four different bias mitigation approaches are explored and evaluated based on data quality, predictive recall, and fairness improvement:

- Synthetic Data Generation using Random Oversampling
- Resampling using SMOTE-Female Only
- Fairness-Aware Reweighting
- Exponentiated Gradient – Demographic Parity (*Fairlearn*)
- Exponentiated Gradient – Equalized Odds (*Fairlearn*)

5.1.1 Synthetic Data Generation using Random Oversampling

Synthetic data generation was performed using a noise-augmented random oversampling approach to address the underrepresentation of the sensitive attribute *Sex* in the training data. This process created an expanded female subset to improve training balance [12]. The primary objective of this approach is to reduce bias arising from representation imbalance and support fairer and more reliable model learning [11].

Implementation requirements and results

A noise-augmented random oversampling technique was implemented by resampling minority group (female) instances with replacement to match the majority group size, and adding controlled Gaussian noise (5% of the feature standard deviation) to numerical features. This process was applied only to the training data to prevent data leakage and ensure unbiased model evaluation. After applying this, the Sex distribution is shown in the following Table 5.1.

Attribute Label	Before Oversampling	After Oversampling
Male (1)	579 samples	579 samples
Female (0)	155 samples	579 samples
Total	734 samples	1158 samples

Table 5.1: Sensitive Attribute Distribution Before and After Random Oversampling

Model performance after Random Oversampling is shown in Table 5.2 and gender wise performance is shown in Table 5.3.

Model	Accuracy	Recall	Precision	F1-score
Logistic Regression	0.87	0.88	0.88	0.88
Random Forest	0.89	0.91	0.90	0.90

Table 5.2: Baseline Model performance after Random Oversampling

Model	Gender	Accuracy	Recall	Precision	F1-score
Logistic Regression	Female	0.89	0.67	0.67	0.67
	Male	0.86	0.90	0.90	0.90
Random Forest	Female	0.92	0.67	0.80	0.72
	Male	0.88	0.93	0.90	0.91

Table 5.3: Baseline model performance after random oversampling by gender

Key Observations:

- The baseline models exhibit significant gender disparities, with female recall considerably lower than male recall in both models (0.50 vs 0.92 in Logistic Regression and 0.67 vs 0.93 in Random Forest).
- After applying random oversampling, overall performance metrics remain relatively stable; however, female recall improves for Logistic Regression (from 0.50 to 0.67) while remaining unchanged for Random Forest.
- Despite this partial improvement, a substantial gender gap persists, indicating that random oversampling does not fully address the imbalance in predictive performance.
- Overall, random oversampling provides limited fairness improvement—particularly benefiting Logistic Regression—but is insufficient as a standalone mitigation strategy for reducing gender-based disparities.

5.1.2 Resampling using SMOTE-Female Only

Although the overall dataset is balanced with respect to the target variable (*HeartDisease*), a subgroup imbalance exists within the female population, where the distribution of positive and negative cases is skewed. This can lead to biased model learning and lower recall for female patients, which is particularly critical in healthcare settings [5, 13]. To address this, *SMOTE* (Synthetic Minority Over-Sampling Technique) was applied selectively to the female subset of the training data rather than the entire dataset. This targeted approach generates synthetic samples for the minority class within females, improving class balance and enabling

the model to learn more representative decision boundaries for this group, thereby helping to reduce gender-based bias without distorting the overall data distribution [11, 12].

Implementation requirements and results

The SMOTE algorithm was implemented using the `imblearn` library with the default setting `sampling_strategy = 'auto'`. The resampling procedure was applied exclusively to the training dataset to prevent data leakage and ensure unbiased model evaluation. Furthermore, SMOTE was selectively applied only to the female subset to address subgroup-level imbalance. After applying SMOTE, the resulting class distribution is presented in Table 5.4.

Group	Class	Before SMOTE	After SMOTE
Female	1	111	111
	0	44	111
Male	1	362	362
	0	217	217
Total	1	473	473
	0	261	328
Total Samples		734	801

Table 5.4: Class Distribution Before and After Female-Only SMOTE

Key Observations:

- After applying SMOTE, the female class distribution becomes perfectly balanced (from 111:44 to 111:111), effectively eliminating internal class imbalance and enabling the model to better learn patterns associated with positive heart disease cases in females.
- Since SMOTE was applied only to the female subset, the male distribution remains unchanged, ensuring that the global data structure is not unnecessarily distorted while still addressing subgroup-level bias.

Performance of the Baseline Model after applying SMOTE for the female subset of the dataset is as shown in Table 5.5.

Model	Gender	Accuracy	Recall	Precision	F1-score
Logistic Regression	Female	0.89	0.83	0.63	0.71
	Male	0.88	0.91	0.92	0.91
Random Forest	Female	0.92	0.83	0.71	0.77
	Male	0.85	0.90	0.88	0.89

Table 5.5: Baseline model performance after Female–only SMOTE

Key Observations:

- Female-only SMOTE improves performance for the female subgroup in both models, particularly increasing recall (Logistic Regression: 0.63, Random Forest: 0.71), thereby reducing the gender disparity, with Logistic Regression showing stronger gains.
- Despite these improvements, a performance gap persists, and a slight trade-off is observed for males—especially in Random Forest (accuracy: 0.85)—indicating that SMOTE partially mitigates bias at the cost of minor reductions in majority-class performance.

5.1.3 Fairness Aware Reweighting

To address gender-based disparities, a fairness-aware cost-sensitive learning approach was applied through instance-level sample reweighting. Higher importance was assigned to the protected subgroup, specifically female patients with positive heart disease outcomes. This increases their contribution to the training loss, enabling the model to better learn patterns for underrepresented cases. The approach aims to reduce false negatives among females, thereby improving the True Positive Rate (TPR). Consequently, it helps minimize the Equal Opportunity gap while preserving the original data distribution. This ensures both fairness and clinical reliability in model predictions [7, 8].

Implementation requirements and results

Reweighting was implemented using the *scikit-learn* library through the `class_weight='balanced'` parameter and custom instance-level `sample_weight`. While class weights adjust importance based on class frequency, additional weights were assigned to emphasize the protected subgroup, specifically female patients with positive heart disease outcomes. This approach preserves the original data distribution while enabling fairness-aware learning [13]. The results highlight improved recall and fairness for the underrepresented female group as shown in Table 5.6.

Model	**Gender**	**Recall–Before Reweighting**	**Recall–After Reweighting**
Logistic Regression	Female	0.50	0.83
	Male	0.92	0.86
Random Forest	Female	0.67	0.83
	Male	0.93	0.92

Table 5.6: Gender-wise Recall Comparison – Before vs After Reweighting

- Logistic Regression shows a substantial improvement in female recall from 0.50 to 0.83 after reweighting, while Random Forest improves from 0.67 to 0.83. This indicates a significant enhancement in detecting heart disease for female patients and a notable reduction in gender disparity across both models.

5.1.4 Exponentiated Gradient - Demographic Parity(Fairlearn)

This in-processing bias mitigation algorithm is implemented using the *Fairlearn* library, which enforces fairness constraints during model training to balance predictive performance and equitable treatment across demographic groups [7, 15]. In the context of this study, it is applied to reduce gender-based disparities, ensuring that the model maintains comparable performance for both male and female patients while minimizing bias in heart disease prediction.

Why we need it?

This method helps prevent systematic disparities across demographic groups in heart disease prediction. Traditional machine learning models primarily focus on optimizing accuracy, which can lead to unequal prediction rates across groups and potentially biased outcomes [14]. The Exponentiated Gradient algorithm with Demographic Parity constraint ensures that each group receives comparable positive prediction rates. In this context, it aims to ensure that both male and female patients have an equal likelihood of being predicted as having heart disease, thereby reducing gender-based bias in model predictions.

Implementation and Result

The fairness–aware modeling approach was implemented using Fairlearn's *ExponentiatedGradient* algorithm with a *Demographic Parity* constraint, applied to Logistic Regression and Random Forest models. The sensitive attribute *Sex* was used to enforce fairness during training. Model performance was evaluated separately for male and female groups using recall, while fairness was assessed using two key measures: Equal Opportunity (EO) gap, based on differences in recall, and Demographic Parity (DP) gap, based on differences in positive prediction rates. A comparative analysis between baseline and fairness-constrained models was conducted as shown in Table 5.7 to examine the trade-off between predictive performance and bias reduction across gender groups.

Model	Type	Recall		Fairness Metrics	
		Female	Male	EO Gap	DP Gap
Logistic Regression	Baseline	0.50	0.92	0.42	0.53
	Fairlearn (DP)	0.83	0.84	0.01	0.04
Random Forest	Baseline	0.67	0.93	0.26	0.56
	Fairlearn (DP)	1.00	0.93	0.07	0.04

Table 5.7: Comparison of gender-wise recall, Equal Opportunity Gap, and Demographic Parity Gap before and after applying Fairlearn (Demographic Parity constraint)

Key Observations:

- The application of Fairlearn with Demographic Parity significantly reduces both the Equal Opportunity (EO) gap and Demographic Parity (DP) gap across models (e.g., Logistic Regression EO gap: 0.42 $\rightarrow$ 0.01, Random Forest: 0.26 $\rightarrow$ 0.06), indicating substantial improvement in fairness.
- This improvement is accompanied by a strong increase in female recall (Logistic Regression: 0.50 $\rightarrow$ 0.83, Random Forest: 0.67 $\rightarrow$ 1.00), while male recall slightly decreases in Logistic Regression. However, the perfect recall of 1.00 in Random Forest suggests possible overcompensation toward the minority group, highlighting a fairness–performance trade-off and potential concerns regarding generalization.

5.1.5 Exponentiated Gradient – Equalized Odds(Fairlearn)

This method operates similarly to the *Exponentiated Gradient* approach but enforces *Equalized Odds* constraints, aiming to achieve equality in both true positive and false positive rates across demographic groups. In the context of this study, it ensures that male and female patients have comparable rates of correct (true positive) and incorrect (false positive) heart disease predictions. This helps mitigate bias arising from gender-based disparities and ensures that the model's decisions are not disproportionately favorable toward any particular group.

Implementation and Results

The Equalized Odds-based fairness approach was implemented using Fairlearn's *ExponentiatedGradient* algorithm, applied to Logistic Regression and Random Forest models. The sensitive attribute *Sex* was used to enforce fairness constraints during training. Model performance was evaluated separately for male and female groups using recall (True Positive Rate), while fairness was assessed through two key metrics: TPR gap and FPR gap, representing differences in true positive and false positive rates across groups. A comparative analysis between baseline and fairness-constrained models was conducted to examine the trade-off between predictive performance and reduction of gender-based bias as shown in Table 5.8.

Model	Type	Recall		Fairness Metrics	
		Female	Male	TPR Gap	FPR Gap
Logistic Regression	Baseline	0.50	0.92	0.42	0.10
	Fairlearn (EO)	0.83	0.86	0.03	0.04
Random Forest	Baseline	0.67	0.93	0.26	0.21
	Fairlearn (EO)	0.67	0.94	0.27	0.21

Table 5.8: Comparison of gender-wise recall, TPR Gap, and FPR Gap before and after applying Fairlearn (Equalised Odds constraint)

- The application of Fairlearn with Equalised Odds significantly reduces both TPR and FPR gaps for Logistic Regression (TPR gap: 0.42 $\rightarrow$ 0.03, FPR gap: 0.10 $\rightarrow$ 0.04), indicating improved fairness. However, Random Forest shows no meaningful change in either TPR or FPR gaps, suggesting that the constraint has a limited impact on this model.

5.2 Bias Mitigation for Other Clinical Features

After identifying gender bias in the dataset, this section focuses on addressing feature-level bias in heart disease prediction. Exploratory analysis, bias detection techniques, and explainable AI methods were used to identify influential and potentially biased attributes. In particular, clinical features such as *FastingBS*, *MaxHR*, *RestingBP*, *Cholesterol*, and *RestingECG*, along with the sensitive attribute *Sex*, were identified as contributing to biased model behavior. These features were found to impose disparities between male and female groups based on Gender-Specific Feature Importance analysis using Logistic Regression.

To mitigate these biases, the study employs fairness-aware technique using *AIF360* (pre-processing methods). This approach aims to reduce bias while preserving clinically meaningful patterns, with some methods focusing on data-level adjustments and others enforcing fairness constraints during model training. Overall, this strategy helps promote equitable model performance across demographic groups without compromising predictive reliability.

5.2.1 Reweighing (AIF360)

The pre-processing reweighting technique inspired by *AIF360* was implemented to mitigate feature-level bias by assigning instance weights prior to model training based on both protected group membership and target class distribution. In this approach, clinically relevant features such as *FastingBS*, *MaxHR*, *RestingBP*, *Cholesterol*, and *RestingECG* were transformed into binary protected attributes using meaningful thresholds. Reweighting factors were computed based on the joint probability of each feature group and target label, and combined multiplicatively to generate a unified sample weight for each instance. These weights were then incorporated during model training using Logistic Regression and Random Forest to enable fairness-aware learning.

Why do we need it?

- Prior analysis revealed disparities in model performance between male and female patients, along with potential bias introduced by clinically influential features.
- Standard models tend to learn dominant patterns from majority groups, leading to reduced sensitivity for underrepresented or high-risk subgroups [9].

- By incorporating reweighting, the model is encouraged to better learn minority patterns, thereby improving recall for female patients and reducing the Equal Opportunity gap, while preserving the original clinical data distribution.

Implementation and Result

The reweighting approach was implemented using a manual formulation inspired by the AIF360 *Reweighing* algorithm, rather than directly using its built-in function. The following Table 5.9 represents the performance results of the Sex variable, illustrating the impact of reweighting on model performance and fairness.

Binary protected attributes were derived from selected clinical features to facilitate feature-level bias mitigation. The feature *FastingBS* was retained in its original binary form, while *MaxHR* was transformed using a median-based threshold to distinguish lower heart rate risk groups. Clinically established thresholds were applied to *RestingBP* ($\geq$ 130 mmHg) and *Cholesterol* ($\geq$ 240 mg/dl) to identify high-risk categories. For *RestingECG*, abnormal conditions were captured using either the one-hot encoded *RestingECG_ST* feature or a fallback encoding. These transformations enable the identification of clinically meaningful subgroups for fairness-aware reweighting while preserving interpretability.

Model	**Type**	**Recall**		**Fairness Metric**
		Female	**Male**	**EO Gap**
Logistic Regression	Baseline	0.50	0.92	0.42
	Reweighted	0.67	0.80	0.13
Random Forest	Baseline	0.67	0.93	0.26
	Reweighted	0.83	0.92	0.09

Table 5.9: Comparison of female and male recall with Equal Opportunity Gap before and after AIF360 style reweighting

Key Observations:

- Reweighting reduces the Equal Opportunity Gap in both models, with Logistic Regression decreasing from 0.42 to 0.13 and Random Forest from 0.26 to 0.09, indicating improved fairness across gender groups.
- This improvement is achieved through both an increase in female recall (Logistic Regression: 0.50 $\rightarrow$ 0.67, Random Forest: 0.67 $\rightarrow$ 0.83) and a slight reduction in male recall, reflecting a balanced adjustment that enhances minority detection while moderating majority performance.

5.3 Female Specific Model

To further address potential bias and improve model reliability across demographic groups, a female-specific model was developed as a targeted bias mitigation strategy. This approach involved training a separate predictive model exclusively on data corresponding to female patients, thereby allowing the model to better capture patterns, risk factors, and feature relationships that are more representative of this subgroup. By isolating the training process to female-specific data, the model reduces the influence of majority-group (male) patterns and mitigates the risk of biased predictions.

Aim of this tailored modeling approach is to enhance predictive fairness and performance for female patients, complementing the fairness-aware global model and contributing to a more equitable and robust decision-support system [3].

5.3.1 Female only Full Model

A female-only full model was developed by isolating the dataset to include only female patients and training a dedicated predictive model on this subset. This approach enables the analysis of feature contributions and relationships that are specific to the female population, which may otherwise be overshadowed in a pooled model dominated by male samples [6]. To ensure that feature importance remains unbiased and not influenced by any specific model, the Chi-square (χ^2) statistical test was employed, and the result is shown in Fig. 5.2.

	Feature	Test Type	p-value
9	ST_Slope	Chi-square	2.016282e-13
1	ChestPainType	Chi-square	4.512634e-11
7	ExerciseAngina	Chi-square	1.323391e-09
4	FastingBS	Chi-square	2.470510e-05
8	Oldpeak	T-test	2.557330e-05
2	RestingBP	T-test	4.344113e-04
0	Age	T-test	6.538414e-04
6	MaxHR	T-test	2.321929e-03
3	Cholesterol	T-test	1.271233e-02
5	RestingECG	Chi-square	7.828935e-01

Figure 5.2: Feature importance using the chi-square test for the female-only model

Unlike model-based feature importance methods such as Logistic Regression or Random Forest, which can inherently reflect the biases present in the trained model, the Chi-square method provides a model-agnostic evaluation of the association between each feature and the target variable. By computing feature importance independently of the predictive model, this approach

reduces the risk of reinforcing model-driven bias and allows for a more objective identification of relevant predictors within the female subset.

Female Model Performance

To evaluate the performance of the female-only model, Stratified K-Fold Cross-Validation with $k = 5$ was employed. This technique was specifically chosen due to its suitability for medical datasets, which often exhibit class imbalance. Unlike standard K-Fold cross-validation, stratified splitting preserves the original class distribution in each fold, ensuring that every subset used for training and validation is representative of the overall dataset.

By maintaining consistent class proportions across folds, Stratified K-Fold provides a more reliable and unbiased estimate of model performance, particularly in sensitive domains such as healthcare, where minority class prediction is critical. This approach reduces the risk of misleading evaluation results that may arise from uneven class distributions.
Performance evaluation of the female dataset is shown in Table 5.10.

Model	Mean Recall	Mean Accuracy	Mean Precision	Mean F1–Score
Logistic Regression	0.76	0.79	0.60	0.65
Random Forest	0.50	0.80	0.60	0.54

Table 5.10: Performance evaluation of the female-only model using stratified k-fold cross-validation

Overall, the female-only full model, combined with Chi-square-based feature selection, enhances interpretability and fairness by ensuring that the identified important features genuinely reflect the underlying data patterns for female patients, thereby contributing to a more equitable and reliable decision-support system.

5.3.2 3–Feature Metabolic Model

Building upon the feature importance analysis illustrated in Fig. 4.11 and Fig. 4.12, a focused three-feature metabolic model was developed to further investigate subgroup-specific predictive behavior. The features *Age*, *Cholesterol*, and *FastingBS* were selected, as they demonstrated relatively higher importance for female patients compared to males across both Logistic Regression and Random Forest analyses. These features are clinically relevant metabolic indicators and are known to play a significant role in cardiovascular risk, particularly in women.

By restricting the model to these three features, the objective was to create a simplified yet interpretable model that emphasizes female-dominant risk factors while reducing the influence of features that may be biased toward male patterns. This approach enables a more targeted evaluation of metabolic risk contributions and helps assess whether a reduced feature space can

still achieve meaningful predictive performance.

The performance of the three-feature metabolic model was evaluated on both the female-only dataset and the pooled dataset, as shown in the Table 5.11 and Table 5.12.

Model	Mean Recall	Mean Accuracy	Mean Precision	Mean F1–Score
Logistic Regression	0.71	0.66	0.70	0.67
Random Forest	0.63	0.60	0.64	0.62

Table 5.11: Performance Evaluation of 3–feature Pooled model using Stratified K-Fold Cross Validation

Model	Mean Recall	Mean Accuracy	Mean Precision	Mean F1–Score
Logistic Regression	0.24	0.75	0.69	0.32
Random Forest	0.32	0.70	0.46	0.35

Table 5.12: Performance Evaluation of 3–feature Female only model using Stratified K-Fold Cross Validation

Key Observation:

- This three-feature model is used as a reference baseline to guide the development and evaluation of the subsequent four-feature model.

5.3.3 4–Feature Model

A four-feature model was developed by extending the metabolic feature set with an additional important feature identified from feature importance analysis. This model improves predictive performance while maintaining interpretability, providing a balance between simplicity and effectiveness.

Two types of four-feature models were considered to evaluate different feature selection strategies. The first, referred to as the *Pushing-4* model, includes *Age*, *Cholesterol*, *FastingBS*, and *ST_Slope*, extending the metabolic feature set with an additional diagnostic variable. ST_Slope was included in this model due to its strong clinical and statistical relevance in heart disease prediction as shown in Fig. 5.2. The second, referred to as the *Top-4* model, consists of *ST_Slope*, *ChestPainType*, *ExerciseAngina*, and *FastingBS*, selected based on female feature importance rankings as shown in Fig. 5.2.

Performance of Pushing–4 Feature Model

The performance of the pushing-4 feature model was evaluated on both the female-only dataset and the pooled dataset, as presented in the Table 5.13 and Table 5.14 below. This comparison

provides insight into the model's effectiveness in capturing female-specific patterns as well as its generalizability across the complete population.

Model	Mean Recall	Mean Accuracy
Logistic Regression	0.89	0.76
Random Forest	0.91	0.76

Table 5.13: Performance Evaluation of Pushing-4 Female only model

Model	Mean Recall	Mean Accuracy
Logistic Regression	0.86	0.82
Random Forest	0.83	0.79

Table 5.14: Performance Evaluation of Pushing-4 Pooled model

Since the pooled model includes both male and female samples, unlike the female-only model, subgroup-specific analysis is required. Therefore, the performance of the pushing-4 feature model was assessed separately for male and female groups within the pooled dataset as shown in Table 5.15. This allows for a better understanding of how the model performs across different subpopulations.

Model	Mean Recall		Mean Accuracy	
	Male	Female	Male	Female
Logistic Regression	0.86	0.89	0.83	0.76
Random Forest	0.83	0.91	0.79	0.76

Table 5.15: Gender Specific Performance Evaluation of Pushing-4 Pooled model

Key Observations:

- Female recall improves significantly in the Pushing-4 model (from lower baseline values to values around 0.90), enabling better detection of heart disease in women.
- A slight decrease in accuracy is observed, particularly for females, reflecting a trade-off in favor of improving recall.
- Overall performance becomes more balanced and reliable, demonstrating that the Pushing–4 approach enhances fairness without substantial loss in predictive capability.

Performance of Pushing–4 Model with SMOTE (Female Subset)

To further investigate the impact of targeted resampling, the Pushing–4 feature model was evaluated on the SMOTE-augmented dataset, where synthetic samples were generated specifically

for the female subgroup. This approach aims to address intra-group class imbalance within the female population and improve the model's ability to learn minority class patterns. The performance results for both Logistic Regression and Random Forest models across male and female groups are presented in Table 5.16.

Model	Gender	Accuracy	Recall	F1-score
Logistic Regression	Male	0.77	0.78	0.82
Logistic Regression	Female	0.68	0.83	0.46
Random Forest	Male	0.79	0.84	0.84
Random Forest	Female	0.68	0.83	0.46

Table 5.16: Performance of Pushing–4 model on SMOTE-augmented dataset

Key Observations:

- Both Logistic Regression and Random Forest achieve identical recall for females (0.83), indicating improved detection of heart disease cases in the minority subgroup after SMOTE augmentation.
- Despite this improvement, a noticeable disparity remains in overall performance, as female accuracy (0.68) and F1-score (0.46) are substantially lower than those for males, suggesting a higher number of false positives for female predictions.
- Random Forest outperforms Logistic Regression for males in terms of accuracy (0.79 vs 0.77) and recall (0.84 vs 0.78), while both models exhibit identical performance for females, indicating a limited model-specific advantage for the minority subgroup.
- Overall, SMOTE improves recall for females but does not translate into balanced performance across metrics, highlighting a trade-off where increased sensitivity leads to reduced precision and F1-score for the minority group.

Performance of Top–4 Feature Model

Similarly, performance of the top-4 feature model was evaluated on both the female-only dataset and the pooled dataset, as presented in the Table 5.17 and Table 5.18.

Model	Mean Recall	Mean Accuracy
Logistic Regression	0.84	0.85
Random Forest	0.80	0.82

Table 5.17: Performance Evaluation of Top-4 Female only model

Model	Mean Recall	Mean Accuracy
Logistic Regression	0.88	0.85
Random Forest	0.87	0.84

Table 5.18: Performance Evaluation of Top-4 Pooled model

Furthermore, performance of Top-4 feature model for male and female groups is shown in the Table 5.19.

Model	Mean Recall		Mean Accuracy	
	Male	Female	Male	Female
Logistic Regression	0.88	0.84	0.85	0.85
Random Forest	0.88	0.80	0.84	0.82

Table 5.19: Gender Specific Performance Evaluation of Top-4 Pooled model

Key Observations:

- Top-4 pooled model demonstrates balanced performance across genders, with recall and accuracy values remaining relatively close for male and female groups.
- A slight disadvantage in recall is observed for females, particularly in the Random Forest model.
- The models maintain consistently high accuracy, suggesting stable predictive performance without significant trade-offs.

Performance of Top–4 Model with SMOTE (Female Subset)

Similarly, the Top–4 feature model was also evaluated using the SMOTE-augmented dataset. This setup enables a focused assessment of how well the selected feature set adapts to improved subgroup representation and whether it enhances predictive performance across gender groups. The resulting performance metrics for Logistic Regression and Random Forest models are presented in Table 5.20.

Model	Group	Accuracy	Recall	F1-score
Logistic Regression	Male	0.85	0.85	0.88
Logistic Regression	Female	0.82	1.00	0.63
Random Forest	Male	0.84	0.88	0.88
Random Forest	Female	0.74	1.00	0.54

Table 5.20: Performance of Top–4 model on SMOTE-augmented dataset

Key Observations:

- The Top–4 model achieves perfect recall (1.00) for females in both Logistic Regression and Random Forest, indicating that all positive heart disease cases are identified; however, this is accompanied by lower F1-scores (0.63 and 0.54), suggesting an increase in false positives.
- Logistic Regression demonstrates more balanced performance across genders, with comparable accuracy (0.85 for males vs 0.82 for females), while Random Forest shows a larger drop in female accuracy (0.74), indicating less stable generalization for the minority subgroup.
- Compared to the Pushing–4 model, the Top–4 model achieves higher overall performance, with improvements in accuracy and F1-score for both genders, while also increasing female recall to 1.00. However, the relatively lower F1-scores for females, particularly in Random Forest, indicate that this gain in recall does not fully translate into balanced predictive performance.

Overall, the Top–4 model demonstrates improved performance and stability under SMOTE compared to the Pushing–4 model, with higher accuracy and F1-scores across both genders and a substantial increase in female recall. However, the presence of perfect recall for females (1.00) alongside comparatively lower F1-scores indicates potential imbalance in predictions. These results suggest that feature selection plays a critical role in enabling models to better utilize synthetic data, while also influencing the trade-off between sensitivity and overall predictive reliability.

6 Evaluation

6.1 Result Comparison

This section compares the performance of all model configurations evaluated in this study. Results are organized into three groups: baseline performance, sensitive-attribute mitigation strategies, and female-specific models. The primary metrics of interest are female recall and the Equal Opportunity (EO) gap, which together capture how effectively each approach addresses the gender disparity identified in the dataset. Baseline Performance of the overall and gender-stratified results for the two classifiers, Logistic Regression and Random Forest, before any mitigation is applied, is shown in Table 6.1.

Model	Accuracy	Recall	F1-Score	Female Recall	Male Recall	EO Gap
Logistic Regression	0.89	0.89	0.90	0.50	0.92	0.42
Random Forest	0.88	0.91	0.89	0.67	0.93	0.26

Table 6.1: Baseline Model Performance

Logistic Regression achieves an overall recall of 0.89; however, female recall drops sharply to 0.50 compared to male recall of 0.92, resulting in a substantial Equal Opportunity (EO) gap of 0.42. In clinical terms, this implies that nearly half of female patients with actual heart disease would be missed by the model, representing a critical false negative risk with potentially severe consequences. The EO gap quantifies this disparity as the difference in true positive rates between male and female groups, highlighting a significant fairness concern.

This gap arises as the dataset contains 725 male records (79%) and only 193 female records (21%). This structural imbalance means that during training, the model is exposed to roughly 3.75 male examples for every female example. As a result, the decision boundary learned by the model is implicitly optimized for male patients. The effect is compounded by the fact that, as shown in Chapter [4], the most predictive features – particularly ST Slope, ChestPainType, and ExerciseAngina – exhibit distinct distributional patterns across genders. A model trained without accounting for this distributional shift will generalize poorly to the underrepresented group.

Random Forest demonstrates slightly better overall performance and a narrower disparity, with female recall of 0.67 compared to male recall of 0.93, yielding an EO gap of 0.26. Although

this represents an improvement over Logistic Regression, the gap remains considerable, indicating that a significant proportion of female cases are still not being correctly identified. These baseline models, therefore, establish the reference point against which all fairness mitigation strategies are evaluated. .

The baseline results also reveal the inadequacy of aggregate metrics in capturing bias. Both models report high overall accuracy (0.89 and 0.88) and recall (0.89 and 0.91), which may create a misleading impression of strong and reliable performance. However, only through gender-stratified evaluation does the underlying disparity become evident, emphasizing the importance of subgroup analysis in high-stakes applications such as healthcare.

6.1.1 Sensitive Attribute Mitigation

Table 6.2 compares the five mitigation strategies applied to address gender imbalance. A dash (–) indicates the metric was not the primary focus of that strategy's evaluation.

Strategy	Model	Recall		EO Gap
		Male	Female	
Baseline	Logistic Regression	0.92	0.50	0.42
	Random Forest	0.93	0.67	0.26
Random Oversampling	Logistic Regression	0.90	0.67	0.23
	Random Forest	0.93	0.67	0.26
Female–only SMOTE	Logistic Regression	0.91	0.83	0.08
	Random Forest	0.90	0.83	0.07
Fairness Reweighting	Logistic Regression	0.86	0.83	0.03
	Random Forest	0.92	0.83	0.09
EG – Demographic Parity	Logistic Regression	0.84	0.83	0.01
	Random Forest	0.93	1.00	0.07
EG – Equalized Odds	Logistic Regression	0.86	0.83	0.03
	Random Forest	0.94	0.67	0.27
AIF360 Style Reweighting	Logistic Regression	0.80	0.67	0.13
	Random Forest	0.92	0.83	0.09

Table 6.2: Sensitive Attribute Mitigation Strategy Comparison

Random Oversampling represents the simplest intervention, where female records are duplicated at random until class balance is achieved. However, the results reveal its limitations: female recall for Logistic Regression improves from 0.50 to 0.67, reducing the EO gap from 0.42 to 0.23, while Random Forest shows no improvement in female recall (remaining at 0.67) and no change in EO gap (0.26). This occurs because random oversampling does not introduce new information, but merely replicates existing observations. As a result, the model learns from repeated patterns without gaining additional insight into female-specific disease characteristics, leading to limited fairness improvement.

This limitation motivates the use of SMOTE (Synthetic Minority Over-Sampling Technique) as a more principled alternative. Unlike random duplication, SMOTE generates synthetic samples through interpolation in feature space. Applying SMOTE selectively to the female subgroup leads to substantial fairness improvements. For Logistic Regression, female recall increases from 0.50 to 0.83, reducing the EO gap from 0.42 to 0.08. Random Forest achieves a similar improvement, increasing female recall from 0.67 to 0.83 and reducing the EO gap from 0.26 to 0.07. These results indicate that targeted resampling enhances minority class detection while maintaining balanced performance, making it significantly more effective than naive oversampling.

Fairness-aware reweighting provides a more theoretically grounded alternative by modifying the loss function rather than the data distribution. For Logistic Regression, female recall improves from 0.50 to 0.83 while male recall decreases from 0.92 to 0.86, reducing the EO gap from 0.42 to 0.03. Random Forest also benefits, with female recall increasing from 0.67 to 0.83 and the EO gap reducing from 0.26 to 0.09. This suggests that reweighting can improve fairness through a balanced adjustment—improving minority detection while slightly moderating majority performance.

Under the Exponentiated Gradient (EG) framework with the Demographic Parity constraint, fairness improves significantly, with EO gaps reduced to 0.01 for Logistic Regression and 0.07 for Random Forest. However, Random Forest achieves perfect female recall (1.00), indicating potential overcompensation toward the minority group. While this eliminates false negatives for females, it may lead to an increase in false positives, raising concerns about practical utility in clinical settings.

In contrast, the Equalized Odds constraint provides a more balanced outcome for Logistic Regression, reducing the EO gap to 0.03 while maintaining strong recall for both groups (0.86 vs 0.83). However, Random Forest shows no improvement under this constraint, with the EO gap remaining high at 0.27. This highlights that fairness interventions can be model-dependent, with Equalized Odds being effective for linear models but less impactful for more complex models.

Finally, AIF360-style reweighting reduces the EO gap in both models, but with different dynamics. For Logistic Regression, the gap decreases from 0.42 to 0.13 due to both an increase in female recall (0.50 to 0.67) and a reduction in male recall (0.92 to 0.80). Random Forest shows a stronger improvement, with female recall increasing to 0.83 and EO gap reducing to 0.09. This indicates that reweighting can improve fairness through both minority uplift and

majority adjustment, rather than relying solely on performance reduction.
Key Observations:

- Random Oversampling provides limited fairness improvement, with female recall improving for Logistic Regression (0.50 $\rightarrow$ 0.67) and EO gap reducing (0.42 $\rightarrow$ 0.23), but no change for Random Forest (female recall remains 0.67, EO gap 0.26), indicating that duplicating samples has limited effectiveness.
- Female-only SMOTE significantly improves fairness by increasing female recall (Logistic Regression: 0.50 $\rightarrow$ 0.83, Random Forest: 0.67 $\rightarrow$ 0.83) and reducing EO gaps (0.42 $\rightarrow$ 0.08 and 0.26 $\rightarrow$ 0.07), demonstrating the effectiveness of synthetic sample generation.
- Fairness-aware reweighting further reduces disparity, with Logistic Regression improving female recall (0.50 $\rightarrow$ 0.83) and reducing EO gap to 0.03, while Random Forest also shows improvement (EO gap: 0.26 $\rightarrow$ 0.09), indicating a balanced adjustment of model predictions.
- Demographic Parity achieves very low EO gaps (0.01 for Logistic Regression, 0.07 for Random Forest), but results in overcompensation, with Random Forest reaching perfect female recall (1.00), raising concerns about increased false positives and reduced practical reliability.
- Equalized Odds provides the best fairness–performance trade-off for Logistic Regression (EO gap: 0.03 with balanced recall), but shows no improvement for Random Forest (EO gap remains high at 0.27), highlighting model-dependent effectiveness.
- AIF360-style reweighting reduces EO gaps in both models (Logistic Regression: 0.42 $\rightarrow$ 0.13, Random Forest: 0.26 $\rightarrow$ 0.09) through a combination of improved female recall and reduced male recall, indicating a balanced but moderate fairness improvement.

6.1.2 Female Specific Models Comparison

Table 6.3 presents the female recall, accuracy, and EO gap for all female-specific model configurations evaluated in both female-only and pooled settings.

The female–only full–feature model highlights a key limitation of the dataset. Training Logistic Regression exclusively on the 193 female records yields moderate performance (recall: 0.76), whereas Random Forest performance deteriorates significantly, with female recall dropping to 0.50. This can be attributed to data scarcity, as only 44 positive cases are available within the female subset, which is insufficient for training robust models—particularly high-capacity models such as Random Forest—without overfitting. Consequently, the model struggles to generalize to unseen data, leading to weaker predictive performance. This finding underscores

Model	Configuration	Male Recall	Female Recall	Accuracy	EO Gap
Logistic Regression	Baseline Model	0.92	0.50	0.89	0.42
Random Forest	Baseline Model	0.93	0.67	0.88	0.26
Logistic Regression	Female-Only Model	–	0.76	0.79	–
Random Forest	Female-Only Model	–	0.50	0.80	–
Logistic Regression	Pushing-4 (Female-Only)	–	0.89	0.76	–
Random Forest	Pushing-4 (Female-Only)	–	0.91	0.76	–
Logistic Regression	Pushing-4 (Pooled)	0.86	0.89	0.82	0.03
Random Forest	Pushing-4 (Pooled)	0.83	0.91	0.79	0.08
Logistic Regression	Pushing-4 (SMOTE)	0.78	0.83	0.76	0.05
Random Forest	Pushing-4 (SMOTE)	0.84	0.83	0.77	0.01
Logistic Regression	Top-4 (Female-Only)	–	0.84	0.85	–
Random Forest	Top-4 (Female-Only)	–	0.80	0.82	–
Logistic Regression	Top-4 (Pooled)	0.88	0.84	0.85	0.04
Random Forest	Top-4 (Pooled)	0.88	0.80	0.84	0.08
Logistic Regression	Top-4 (SMOTE)	0.85	1.00	0.86	0.15
Random Forest	Top-4 (SMOTE)	0.88	1.00	0.88	0.12

Table 6.3: Model performance comparison across configurations with gender-specific evaluation

the severe underrepresentation of female patients (21%) and highlights the inherent limitations of female-only modeling approaches in this dataset.

The Pushing–4 and Top–4 feature models adopt a different strategy by refining the feature space rather than restricting the training population. The Pushing–4 model (Age, Cholesterol, FastingBS, ST_Slope) combines metabolic and diagnostic indicators, while the Top–4 model (ST_Slope, ChestPainType, ExerciseAngina, FastingBS) focuses on the most predictive features identified for female outcomes. These feature sets aim to improve signal quality and reduce noise in model learning.

When trained on pooled data, both feature-reduced models demonstrate strong performance and improved fairness. The Pushing–4 model achieves high female recall (0.89 for Logistic Regression and 0.91 for Random Forest), matching or exceeding the performance of sensitive–attribute mitigation strategies, while maintaining low EO gaps (0.03 and 0.08), indicating balanced predictions across genders. Notably, this improvement is achieved without altering the data distribution, demonstrating that carefully selected features can provide stronger and more balanced predictive signals across genders. This highlights feature selection, guided by domain knowledge, as a powerful and interpretable bias mitigation strategy.

The Pushing–4 pooled Logistic Regression model achieves the lowest EO gap across all configurations (0.03), indicating the most balanced performance between male and female recall. In contrast, the Top–4 pooled Logistic Regression model achieves a slightly higher EO gap (0.04) with a marginally lower female recall (0.84 compared to 0.89 for the Pushing–4 model).

This trade–off reflects differing operational priorities: in screening scenarios, where minimizing false negatives is critical, the Pushing–4 model may be preferable due to its higher female recall. Conversely, in contexts where maintaining a balance between sensitivity and overall performance is important, the Top–4 model provides a competitive and stable alternative.

In contrast, the application of SMOTE-based augmentation reveals mixed effects when combined with reduced feature spaces. For the Pushing–4 model, both Logistic Regression and Random Forest achieve relatively balanced recall across genders (Logistic Regression: 0.78 vs 0.83, Random Forest: 0.84 vs 0.83), resulting in very low EO gaps (0.05 and 0.01). However, this improvement in fairness is accompanied by a slight reduction in overall accuracy compared to pooled models, indicating a trade-off between balance and overall predictive performance.

For the Top–4 model, SMOTE leads to a different behavior. Both Logistic Regression and Random Forest achieve perfect female recall (1.00), but this is accompanied by higher EO gaps (0.15 and 0.12), as male recall remains lower. This behavior is indicative of potential overfitting and overcompensation toward the minority group, where the model becomes highly sensitive to female cases at the expense of balanced predictions. While this may be desirable in high-risk screening scenarios, it raises concerns about stability and the potential increase in false positives.

Overall, these findings demonstrate that feature selection is a more stable and interpretable bias mitigation strategy compared to synthetic data augmentation. While SMOTE can improve subgroup recall, its effectiveness is highly dependent on the feature space and may introduce trade-offs in model balance and generalization.

Key Observations:

- Female–only training improves Logistic Regression recall ($0.50 \rightarrow 0.76$) but leads to weaker performance in Random Forest (recall $= 0.50$), highlighting the limitations imposed by data scarcity.
- The Pushing–4 model achieves the highest female recall (0.89 for Logistic Regression and 0.91 for Random Forest) across pooled configurations, demonstrating the effectiveness of targeted feature selection.
- The pooled Pushing–4 model maintains high recall while achieving stable generalization, with low EO gaps (0.03 and 0.08) and strong overall accuracy (0.82 and 0.79).
- The Top–4 model provides stable and balanced performance, with slightly lower female recall but consistent accuracy and moderate EO gaps across both classifiers.

- The Pushing–4 pooled Logistic Regression model achieves the lowest EO gap (0.03), representing the best fairness–performance trade-off among all configurations.
- SMOTE-based augmentation for the Pushing–4 model improves female recall (0.50 $\rightarrow$ 0.83) and achieves very low EO gaps (0.05 and 0.01), but at the cost of reduced overall accuracy, indicating a trade-off between fairness and performance.
- For the Top–4 model, SMOTE leads to perfect female recall (1.00) for both classifiers, suggesting overcompensation toward the minority group and potential instability in predictions.
- Random Forest remains relatively stable under SMOTE in the Pushing–4 configuration, but shows increased imbalance in the Top–4 setting due to the rise in EO gap.
- Feature selection proves to be a more reliable and stable bias mitigation strategy compared to data-level augmentation, as it consistently improves fairness while maintaining strong predictive performance.

6.2 Analysis

The results across all methodologies converge on a coherent picture that goes beyond numerical performance to showcase the structural problem of gender bias in cardiovascular AI.

6.2.1 How Bias Manifests in this Dataset

Gender bias in this study operates at multiple levels simultaneously, and it is important to distinguish between them to understand why an individual mitigation strategy can or cannot address.

At the data level, the most visible manifestation is the 79% – 21% male-to-female ratio. This imbalance means that the training data that the model uses to build its decision boundary is dominated by male patient profiles. But the imbalance in sample count alone does not fully explain the performance gap. The EDA results in Chapter [4] show that key predictive features differ meaningfully in their distributions across genders. The model, therefore, learns a decision boundary calibrated to the more common, male-typical presentation of the disease.

At the feature level, the bias is not merely about quantity but about which signals are most informative for which group. The gender-stratified feature importance analysis reveals that the relative contribution of features differs between male and female patients. A single model trained on pooled data learns a generalized feature weighting that reflects the majority group's

signal more strongly than the minority group's. This is why feature–guided modelling – selecting features that are specifically informative for female patients – yields such significant improvements in female recall.

6.2.2 How Bias Can Be Reduced: A Comparative Assessment

The results allow for a structured comparison of bias reduction strategies according to their mechanism, effectiveness, and practical implications.

Pre-processing strategies that target data composition – random oversampling in this study – are the most computationally straightforward but the least effective. The fundamental reason is that they do not address the feature-level distributional asymmetry described above. SMOTE, which generates synthetic minority-class samples by interpolating in feature space between real neighbors, offers a more principled alternative. By creating new data points that expand the effective feature coverage of the minority class rather than simply duplicating existing ones, SMOTE can better represent the range of female clinical presentations.

Fairness–aware reweighting and in-processing constraints address the algorithmic level of bias. They steer the model toward decision boundaries that are better calibrated across groups. The Equalized Odds constraint achieves this most effectively because it targets both TPR and FPR parity simultaneously, preventing the overcompensation that the Demographic Parity constraint produces. The advantage of these methods is that they leave the dataset unchanged, making them applicable when data augmentation is not feasible.

Feature–guided modelling addresses the feature level of bias most directly. By identifying features that carry a stronger and more balanced predictive signal for female patients – a process that requires clinical domain knowledge in addition to statistical analysis. The Pushing–4 and Top–4 models achieve fairness improvements that are both large and interpretable. Unlike algorithmic constraints, which are opaque to clinical practitioners, a feature–guided model can be directly validated against clinical knowledge about female CVD presentation: ST Slope, for example, is an informative indicator of exercise–induced ischaemia, which presents differently in women than in men.

Whereas, when Feature–guided modeling is considered alongside SMOTE-based augmentation, it provides additional insight into the role of feature representation in bias mitigation. While SMOTE aims to improve minority representation by generating synthetic samples, its effectiveness is inherently dependent on the quality and structure of the feature space. In cases where features are carefully selected based on clinical relevance and subgroup-specific importance,

such as in the Pushing–4 and Top–4 models, the reliance on synthetic data becomes less critical. Feature-guided approaches inherently enhance the model's ability to capture meaningful patterns for underrepresented groups, thereby reducing bias without altering the original data distribution.

In contrast, applying SMOTE on suboptimal or reduced feature spaces may introduce artificial patterns that do not generalize well, potentially affecting model stability. Therefore, feature selection and data augmentation should not be treated as independent strategies, but rather as complementary approaches.

The comparative analysis of the applied bias mitigation strategies indicates that no single technique is sufficient to address all forms of bias within the model. Data-level approaches, such as oversampling, improve representation but do not account for biases introduced during model learning. Similarly, algorithmic techniques, including reweighting and fairness constraints, help adjust model behavior but cannot fully compensate for imbalances or missing patterns in the feature space. Feature-level interventions, while effective in highlighting clinically relevant patterns, rely on domain knowledge and may not generalize across different datasets.

6.2.3 Analysis Outcome:

The results show that most mitigation approaches reduce the gender disparity present at baseline, though through different mechanisms and with varying trade-offs. Logistic Regression benefits more consistently from mitigation strategies, showing substantial improvements in female recall and reductions in EO gap across multiple configurations. In contrast, Random Forest is relatively more balanced at baseline but exhibits mixed responses to mitigation, particularly under female-only training and certain SMOTE configurations, where performance gains are less consistent.

Among the mitigation strategies, Fairness Reweighting emerges as the most consistent and reliable approach, achieving strong improvements in female recall (up to 0.83) while reducing EO gaps across both models (0.03 for Logistic Regression and 0.09 for Random Forest). Female–only SMOTE follows as an effective data-level technique, significantly improving female recall (0.83) and reducing EO gaps (0.08 and 0.07), demonstrating strong performance across both models without introducing extreme imbalance. Equalized Odds provides comparable improvements for Logistic Regression, increasing female recall to 0.83 and reducing the EO gap to 0.03; however, its effectiveness is model-dependent, as Random Forest shows no improvement and even a higher EO gap (0.27). Demographic Parity ranks next, achieving the lowest EO gap (0.01 for Logistic Regression), but at the cost of overcompensation—particularly in Random Forest, where female recall reaches 1.00, indicating potential imbalance in predictions. Finally,

AIF360-style Reweighting provides only moderate improvement, and in the case of Logistic Regression reduces female recall (to 0.67), making it the least effective as a standalone mitigation strategy and better suited as a complementary technique.

The feature–based models demonstrate that gender-guided feature selection can achieve strong fairness outcomes without requiring algorithmic constraints. The Pushing–4 pooled model produces the highest female recall (0.89 for Logistic Regression and 0.91 for Random Forest) with low EO gaps (0.03 and 0.08) and strong overall accuracy (0.82 and 0.79). In comparison, the Top–4 pooled Logistic Regression model achieves a slightly more balanced performance (EO gap $= 0.04$), though with lower female recall (0.84). These findings confirm that incorporating clinically relevant, female-sensitive features into the model design is an effective and interpretable approach to reducing gender bias in cardiovascular disease prediction.

When evaluated in combination with SMOTE-based augmentation, the feature-based models highlight the importance of feature representation in determining the effectiveness of synthetic data techniques. While SMOTE improves subgroup balance, its impact varies across feature configurations. In the Pushing–4 setting, SMOTE leads to more balanced recall across genders (approximately 0.83) with very low EO gaps (0.05 and 0.01), but this improvement comes with a reduction in overall accuracy compared to pooled models, indicating a trade-off between fairness and predictive performance.

In contrast, the Top–4 model exhibits different behavior under SMOTE, achieving perfect female recall (1.00) for both classifiers but with higher EO gaps (0.15 and 0.12), reflecting overcompensation toward the minority group. This suggests that while SMOTE can significantly improve sensitivity, it may also introduce imbalance depending on how the feature space interacts with synthetic samples. Overall, these observations reinforce that the effectiveness of SMOTE is highly dependent on feature representation. While it can improve subgroup performance, feature-guided modeling, particularly the Pushing–4 configuration, provides a more stable and reliable foundation for bias mitigation without introducing extreme trade-offs.

7 Conclusion and Future Work

7.1 Conclusion

This research investigated gender-specific bias in cardiovascular disease (CVD) prediction models, addressing a clinically significant gap in the fairness of AI–driven healthcare systems. Motivated by the well-documented underrepresentation of women in cardiac research and the consequent skewing of predictive algorithms toward male-centric patterns, this study systematically examined bias sources, quantified their impact, and evaluated mitigation strategies. The Heart Failure Prediction Dataset, comprising 918 patient records drawn from five publicly available sources, was used throughout the study. A key structural finding was the severe demographic imbalance in the sensitive attribute Sex, with male instances (725, 79%) significantly outnumbering female instances (193, 21%). This imbalance, as confirmed through exploratory data analysis, uni-variate, bi-variate, and group disparity analyses, formed the primary source of gender bias permeating the entire modeling pipeline.

7.1.1 Bias detection Findings

The baseline classification results confirm the presence of gender bias across the evaluated classifiers. The most pronounced disparity is observed in Logistic Regression, where female recall (0.50) falls substantially below male recall (0.92), resulting in an EO gap of 0.42. Clinically, this implies that nearly half of female patients with heart disease would be misclassified as healthy, leading to a significantly elevated risk of missed diagnosis and delayed treatment. These findings highlight the critical need for bias mitigation, particularly in high-stakes healthcare applications where such disparities can have severe consequences.

Exploratory data analysis identified the primary driver of this disparity: the dataset's severe demographic imbalance (79% male, 21% female). This structural imbalance propagates through the modeling pipeline in two distinct ways. First, the training signal is quantitatively dominated by male patient profiles, biasing the learned decision boundary. Second, and more subtly, the most predictive features exhibit different distributions between genders – so even if the sample counts were balanced, a model trained without gender–stratified analysis would still tend to learn a feature weighting calibrated to the majority group's disease presentation.

Gender–stratified feature importance analysis was particularly informative. Mutual information analysis identified Oldpeak and MaxHR as the most predictive numerical features overall, while Cramér's V analysis highlighted ST_Slope, ChestPainType, and ExerciseAngina as the strongest categorical predictors. Critically, the relative ranking of these features differed between male and female subgroups – demonstrating that the disease manifests through partially different clinical signals in the two populations. This finding has direct implications for model design: a single, pooled feature set will inevitably be more inclined towards the majority group's signal.

7.1.2 Bias Mitigation Outcomes

Six mitigation strategies were applied to address the imbalance in the sensitive attribute *Sex*. Random Oversampling produced limited improvement, increasing female recall for Logistic Regression ($0.50 \rightarrow 0.67$) and reducing the EO gap ($0.42 \rightarrow 0.23$), while showing no improvement for Random Forest. This result is theoretically consistent, as duplicating minority samples does not expand the feature–space representation, limiting the model's ability to learn new patterns for female patients.

To address this limitation, SMOTE (Synthetic Minority Over-Sampling Technique) was applied to generate synthetic female samples. Unlike random duplication, SMOTE improves feature–space coverage by interpolating between minority instances. This resulted in a substantial improvement in female recall ($0.50 \rightarrow 0.83$ for Logistic Regression and $0.67 \rightarrow 0.83$ for Random Forest), along with a reduction in the EO gap (to 0.08 and 0.07), demonstrating its effectiveness in enhancing minority-group representation and detection.

Fairness-Aware Reweighting provided a more targeted, algorithm-level intervention by increasing the loss associated with female misclassification. For Logistic Regression, this raised female recall from 0.50 to 0.83 while reducing the EO gap from 0.42 to 0.03, without altering the data distribution. Random Forest also showed improvement, with female recall increasing from 0.67 to 0.83 and the EO gap reducing to 0.09, indicating that reweighting can achieve balanced fairness improvements across models.

Among in–processing methods, the Exponentiated Gradient (EG) framework showed contrasting outcomes depending on the constraint applied. The Demographic Parity constraint achieved perfect female recall (1.00) for Random Forest, and very low EO gap for Logistic Regression (0.01), but this is indicative of overcompensation, potentially increasing false positives and reducing clinical reliability. In contrast, the Equalized Odds constraint provided a balanced improvement for Logistic Regression, reducing the EO gap to 0.03 while maintaining female recall at 0.83. However, its effectiveness is model-dependent, as Random Forest showed no improvement and a higher EO gap (0.27). This highlights that while Equalized Odds can achieve

a strong fairness–performance trade–off, its benefits may not generalize across model types.

Finally, AIF360-style Reweighting reduces the EO gap through a combination of lowering male recall and modestly improving female recall. For Logistic Regression, female recall increases to 0.67 while male recall decreases, reducing the EO gap to 0.13. Random Forest shows a greater improvement, with female recall reaching 0.83 and the EO gap reducing to 0.09. This indicates that while the method achieves moderate fairness gains, it does so through a balanced adjustment rather than a substantial improvement in minority-group learning, limiting its effectiveness as a standalone strategy.

In addition to these mitigation strategies, female-specific feature models produced the most practically compelling results. The Pushing–4 pooled model achieved the highest female recall (0.89 for Logistic Regression and 0.91 for Random Forest) with low EO gaps (0.03 and 0.08), while the Top–4 pooled Logistic Regression model achieved a similarly balanced outcome with a low EO gap (0.04).
When evaluated alongside SMOTE-based augmentation, the feature-based models exhibit configuration-dependent behavior rather than instability. The Pushing–4 model achieves balanced recall with low EO gaps but at a slight cost to overall accuracy, whereas the Top–4 model shows overcompensation, with perfect female recall (1.00) and higher EO gaps. This indicates that synthetic data alone does not guarantee improved fairness and may introduce trade-offs depending on the feature space, reinforcing that carefully selected features provide a more reliable and stable approach to bias mitigation.

These findings highlight that feature selection informed by domain knowledge can effectively complement fairness-aware techniques to achieve balanced and reliable model performance.

7.1.3 Key Takeaways:

The findings of this research collectively establish several important conclusions for the field of fair machine learning in healthcare:

- Gender bias is clearly measurable in cardiovascular disease datasets. The under-representation of women in the Heart Failure Prediction Dataset translates into significantly lower recall for female patients in baseline models, highlighting a critical fairness concern in clinical prediction.
- No single mitigation strategy consistently dominates across all models and metrics. Each approach involves trade-offs between recall improvement, fairness gap reduction, and overall accuracy, and therefore must be selected based on the specific clinical objective.

- Feature-guided modeling emerges as a stable and interpretable mitigation approach. The Pushing–4 and Top–4 models demonstrate that incorporating clinically relevant, female-sensitive features can substantially improve fairness without requiring complex algorithmic constraints.
- Fairness and clinical utility can be jointly improved. Several mitigation strategies achieve meaningful reductions in gender disparity while maintaining competitive overall performance, indicating that fairness improvements do not necessarily require significant sacrifices in predictive accuracy.

In summary, this research demonstrates that gender-aware machine learning is both necessary and achievable in cardiovascular disease prediction. By combining rigorous bias detection, diverse mitigation strategies, and targeted female-specific modeling, it is possible to develop predictive systems that support the responsible and equitable deployment of AI in clinical decision-making.

7.1.4 Answers to Research Questions

Answer to RQ1:

How can gender–bias be formally defined and measured in cardiovascular disease prediction models with respect to the given dataset?

Bias in this project is formally defined as a disparity in model performance between different groups (male and female patients). It is measured using:
Recall difference (TPR disparity) – Difference in true positive rates across genders
Equal Opportunity Gap (EO Gap) – Gap in sensitivity between male and female
Demographic Parity Gap (DP Gap) – Difference in positive prediction rates
Equalized Odds (FPR Gap) – Difference in false positive rates (for Equalized Odds)

The results show that baseline models exhibit higher recall for male patients than female patients, indicating the presence of gender-based bias in the dataset and model predictions.

Answer to RQ2:

How effective are bias mitigation techniques in reducing gender-based disparities without significantly compromising overall model performance?

Bias mitigation techniques in this study demonstrated varying levels of effectiveness in reducing gender-based disparities while maintaining overall model performance.

Fairness-constrained approaches showed mixed outcomes. Equalized Odds achieved substantial improvement for Logistic Regression by reducing disparity in true positive rates, resulting in more balanced predictions with minimal impact on performance. However, its effectiveness was model-dependent, as Random Forest showed no improvement and even increased disparity. In contrast, the Demographic Parity constraint reduced disparity further but introduced overcompensation, particularly by inflating female recall, indicating a trade-off between fairness and prediction reliability.

Reweighting techniques provided consistent and balanced improvements across models. Fairness-aware reweighting significantly improved female recall while reducing the EO gap without modifying the data distribution, making it one of the most reliable approaches. AIF360-style reweighting also reduced disparity, though with more moderate gains, indicating its suitability as a complementary method rather than a primary solution.

Feature-based approaches (e.g., Pushing–4 and Top–4) achieved strong fairness improvements by increasing female recall while maintaining competitive accuracy. This suggests that part of the observed bias is feature-driven, and that incorporating clinically relevant features can improve fairness without requiring complex constraints. However, when combined with data augmentation techniques such as SMOTE, the results became more variable, indicating that the effectiveness of such methods depends on the underlying feature representation.

7.2 Limitations

While the findings of this study are encouraging, several limitations must be acknowledged in order to contextualize the results appropriately and guide future research.

- The Heart Failure Prediction Dataset contains only 193 female records (21% of the total). This small sample size constrains the reliability of female-specific model training and evaluation. The relatively weaker performance of Random Forest on the female-only subset (recall: 0.50 on the full feature set) reflects the challenges of training complex models on limited data. Conclusions drawn from female-only experiments must therefore be interpreted with caution.
- All experiments were conducted on a single, publicly available dataset compiled from five legacy heart disease repositories. Although widely used in the literature, its composition may not fully capture the demographic and clinical diversity of real-world populations, limiting the generalizability of the findings.
- In certain configurations, particularly under SMOTE-based augmentation, the models achieved perfect female recall (1.00). While this ensures that all positive female cases are

detected, it may indicate overcompensation toward the minority group and a potential increase in false positives. Such behavior can reduce the practical reliability of the model in clinical settings, where excessive false alarms may lead to unnecessary diagnostic procedures. This highlights the need for evaluating additional metrics and validating model performance on external datasets.

- The effectiveness of bias mitigation strategies in this study was found to be model-dependent. Techniques such as Equalized Odds significantly improved fairness for Logistic Regression but showed limited or even adverse effects for Random Forest. This indicates that fairness interventions do not generalize uniformly across model types, and their impact must be evaluated in conjunction with model characteristics.

- The effectiveness of both data-level and algorithmic mitigation techniques was highly sensitive to feature representation. The performance differences observed between Pushing–4 and Top–4 models indicate that the choice of features can significantly influence fairness outcomes. This suggests that bias is not only data-driven but also feature-dependent, and inappropriate feature selection may limit the effectiveness of mitigation strategies.

- Several mitigation strategies improved fairness metrics such as recall and EO gap, but introduced trade-offs in overall model behaviour. In particular, configurations achieving very high female recall were associated with potential imbalance in predictions, raising concerns about false positives and clinical reliability. This highlights the challenge of balancing fairness with practical utility in healthcare applications.

7.3 Future Work

The findings and identified limitations of this study provide a foundation for several promising directions for future research.

Large and More Diverse Datasets

A primary priority for future work is the acquisition and use of larger, more demographically balanced cardiovascular datasets. Datasets that include greater female representation, as well as richer demographic metadata, would enable more statistically robust subgroup–specific analyses and more reliable female-only model training.

Intersectional Fairness Analysis

Future studies should extend the fairness analysis beyond the single sensitive attribute of gender to consider intersectional biases arising from the interaction of multiple protected characteristics. Tools such as multi-attribute fairness metrics and intersectional subgroup analysis frameworks

should be applied to identify and mitigate compound biases that may remain invisible under single-attribute analyses.

Deep Learning

This study was confined to classical machine learning classifiers. Future work should investigate whether deep learning architectures, particularly those with attention mechanisms, offer advantages in capturing gender-specific patterns in cardiovascular data. Attention–based models may be particularly suitable because they can learn feature weighting dynamically, potentially allocating greater weight to female-relevant features without requiring explicit feature pre-selection.

Longitudinal and Symptom-Aware Modelling

The dataset used in this study captures only a few clinical features and does not encode symptom presentation. Future work should incorporate longitudinal patient data, including symptom trajectories over time and temporal biomarker trends, to build models that are more sensitive to female-specific disease progression patterns. Natural language processing techniques applied to clinical notes could also extract richer symptom-level information that is not captured in structured tabular datasets.

7.4 Closing Remarks

Cardiovascular disease continues to be the leading cause of death worldwide for both men and women. The growing integration of machine learning into clinical decision support systems creates both an opportunity and a responsibility: an opportunity to improve early detection and personalised risk stratification, and a responsibility to ensure that these systems do not perpetuate or exacerbate existing inequalities in healthcare.

This research demonstrates that gender bias in CVD prediction is not merely a theoretical concern but a measurable and clinically relevant issue. The findings show that such disparities can be meaningfully reduced through a combination of fairness-aware training, principled feature selection, and subgroup-specific modelling, while maintaining competitive overall performance. In particular, the Pushing–4 and Top–4 feature-based models provide practical and interpretable approaches toward improving gender equity in cardiovascular risk prediction.

The work presented here is a step toward a broader vision: a future in which AI–powered diagnostic tools perform with equal reliability for all patients, regardless of gender, age, or demographic background. Realising this vision will require sustained collaboration between machine learning researchers and clinical practitioners – but the technical foundations demonstrated in this research provide a meaningful and actionable starting point.

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